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IFNN: Enhanced interpretability and optimization in FNN via Adam algorithm

Paulo Vitor de Campos Souza *, Mauro Dragoni

Intelligent Digital Agents Research Group-Fondazione Bruno Kessler Trento, Italy

A R T I C L E I N F O

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A B S T R A C T

This investigation seeks to reconcile the interpretability of artificial intelligence with the imperative of maintaining optimal performance, through the introduction of the IFNN model. This model represents a sophisticated iteration of the Fuzzy Neural Network (FNN) paradigm, engineered for enhanced interpretability. At the heart of this innovation lies the application of the Adam optimization algorithm, integrated into a tri-layered architectural construct. This approach is deliberately designed to elevate the models' accuracy across an array of datasets, thereby positioning the IFNN model as an example of an interpretable AI system that does not compromise on precision.

A seminal aspect of this model is its capacity to transmute logical neurons within the intermediary layer into clear fuzzy rules. This transformative process propels the model beyond the confines of traditional AI frameworks, ushering in a new era of transparency in the AI decision-making arena. Such progress is achieved through a meticulous fuzzy rule-based examination, anchored by an exhaustive appraisal of interpretability metrics, including but not limited to sensitivity, completeness, and the analysis of fuzzy rule consequents. These logical neurons, the progenitors of the aforementioned fuzzy rules, endow the model with the ability to engage in deep, interpretable analyses of data.

The IFNN model has been tested through statistical analysis, interpretability assessments, and empirical validation against real-world datasets about sepsis identification, showcasing its unparalleled ability to unlock and articulate the complex knowledge embedded within data. This model represents a significant evolution in AI methodologies, providing a clear window into the rationale underpinning its decisions, achieved through an advanced fuzzy rule-based methodology and a full spectrum of interpretability metrics.

1. Introduction

In recent years, the proliferation of artificial intelligence (AI) across various sectors of society has underscored the transformative potential of these technologies in addressing complex problems, ranging from healthcare diagnostics to financial forecasting and beyond. Despite their widespread adoption, a significant challenge persists in the form of the "black box" nature of many AI models, which obscures their decision-making processes and limits their interpretability. As the demand for transparency and understand-

* Corresponding author.
E-mail addresses: pdecampossouza@fbk.eu (P.V. de Campos Souza), dragoni@fbk.eu (M. Dragoni).

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ability in AI systems grows, there is an urgent need for models that not only perform with high accuracy but also offer insights into their operational mechanics [1].

Fuzzy Neural Networks (FNNs) represent a notable stride in this direction, merging the interpretability of fuzzy logic with the learning capabilities of neural networks. FNNs are characterized by their layered architecture and the ability to refine knowledge through training, thus embodying a blend of human-like reasoning and machine efficiency. Their composition involves fuzzification layers, logic neuron layers, and optimization techniques, making them adept at handling uncertainty and imprecision, a common occurrence in real-world data [2].

In the past two decades, a multitude of neuro-fuzzy techniques have emerged, differing in their approaches to extracting neurons/rules from data and updating parameters in the antecedent space and architecture [3]. Recent comprehensive surveys on FNN discussed in [4] and [2], explore dozens of approaches, detailing their properties, characteristics, and comparisons regarding parametrization issues and computational demands.

FNNs stand out by harmonizing the robust learning capabilities of neural networks with the nuanced interpretability of fuzzy logic, presenting a number of defining features [5], i.e., the adapted learning flexibility; the interpretability and rule extraction; the integration and hybridization.

These three pillars underscore the versatility and potential of FNNs in tackling the dual challenges of performance and understandability in AI, positioning them as a compelling choice for applications requiring both sophisticated modeling and clarity of reasoning [6].

Fuzzy rules are rigorously constructed with designated levels of significance, positioning them as essential elements for interpretation by underscoring each rule's pivotal role within the overarching rule base. An additional prevalent strategy entails the assessment of a fuzzy rule's pertinence through a methodical diminution throughout its learning trajectory. Within this framework, rules bearing minimal weight—considered to be of lesser importance—may either be methodically trimmed or re-engaged as circumstances require [7].

The evaluation of a rule's relevance is crucial for the FNNs' interpretability. This critical analysis enables the effective handling of pruning criteria and the refinement or modification of the rule set [8].

In the realm of FNNs, the concept of rule utility has been explored to assess the importance of fuzzy rules, as seen in the studies by Wu et al. [9] and Slawinski et al. [10]. Utility is determined by the average normalized activation levels of the rules, offering a quantitative measure of each rule's contribution. A notable example of this approach is the work by Mendel and John [11], which utilizes a statistical method to assess the impact of fuzzy rules on the system's output.

Conventional criteria for evaluating the significance of fuzzy rules, such as their age, density, and statistical impact, have been discussed in various studies, including those by Wang and Mendel [12] and Buckley and Hayashi [13].

A notable direction in FNN research is the simplification of models to improve interpretability. This involves strategies for merging similar neurons or rules, thereby reducing the complexity of the model while retaining its predictive capabilities, as demonstrated in the approaches by Tan et al. [14] and Zhou et al. [15].

Furthermore, the traditional construction of FNNs often relies on t-norms or multi-dimensional kernels, leading to primarily "AND" connections in rule antecedents. This limitation restricts the system's ability to incorporate "OR" connections, which can significantly enhance the interpretability and flexibility in representing class distinctions through rules [16].

1.1. Our approach

This study presents the IFNN, an advanced fuzzy neural network model enhancing interpretability and decision-making accuracy through the integration of fuzzy rules and logical neurons, expanding upon the foundational work by de Campos Souza ([17]). Central to this model are three layers: an Input Layer for fuzzification of crisp inputs into fuzzy values, a Hidden Layer employing logical fuzzy neurons for processing these inputs based on fuzzy logic and empirical or expert-derived rules, and an Output Layer where fuzzy deductions are converted back to crisp outputs, with the Adam optimizer refining neural network weights for improved accuracy.

Significantly, the introduction of the Adam optimizer in the training regimen marks a substantial improvement in the model's precision and adaptability. The model is evaluated across key performance metrics to gauge its effectiveness, interpretability, and alignment with data patterns and domain expertise, ensuring it not only meets conventional performance standards but also excels in delivering meaningful insights through fuzzy rule-based reasoning.

In essence, the IFNN model, with its three-layer structure exemplifies the application of fuzzy logic to augment neural network interpretability and decision-making capability. Our contributions are meticulously integrated within the model's framework, offering a comprehensive solution that significantly advances the interpretability of FNNs:

- **Innovative Neuron Design:** Our model introduces a unique configuration of fuzzy logic neurons, enabling a dynamic interpretation of fuzzy rules. Unlike traditional neurons that are rigid in their logical operations, our model's neurons can adaptively switch between AND and OR logic operations. This flexibility allows for a nuanced representation of data patterns, accommodating a broader spectrum of relationships within the dataset. The adaptability of these neurons facilitates a more expressive rule base, enhancing the model's ability to capture complex, nonlinear relationships inherent in real-world data.
- **Interpretability through Rule Evaluation:** A cornerstone of our model is its emphasis on rule interpretability, achieved by incorporating advanced metrics such as similarity, distinguishability, consistency, and rule activation levels. By evaluating the similarity and distinguishability among the fuzzy rules, our model ensures that each rule contributes uniquely to the decision-making process, thereby enhancing the system's transparency. Moreover, the visualization of rule activation levels across data samples offers insights into the model's reasoning, making it accessible and interpretable to users.

- **Adaptive Learning with Optimized Weights:** The model leverages optimization techniques, notably the Adam optimizer, for the adaptive tuning of neuron weights in its output layer. This approach not only optimizes the model's performance but also ensures that the weights reflect the significance of each input feature and rule, contributing to the overall accuracy and reliability of the system. The integration of these optimization methods marks a significant improvement over traditional training algorithms, offering a more dynamic and responsive learning process.

The primary contribution of this work is the incorporation of the Adam optimization algorithm into Fuzzy Neural Networks (FNN), which facilitates a dynamic and adaptive learning process by the rules' consequents. This innovation resolves the long-standing challenge of interpretability in artificial intelligence without compromising performance. It allows the model not only to maintain high accuracy but also enhances the transparency and understandability of its decision-making processes.

Furthermore, the development and explicit formulation of fuzzy rules within our model provides a clear insight into its operational mechanics. By dynamically adjusting the rules' consequents through the Adam optimization process, our model adapts in real-time to new data, ensuring robustness and versatility across various scenarios. This adaptability and explicit rule formulation ensures that both the architecture and the extracted knowledge can be thoroughly analyzed for informed decision-making, bridging the gap between advanced computational techniques and practical usability in complex scenarios.

Our model's unique approach to rule generation, optimization, and the emphasis on interpretability measures such as similarity, distinguishability, and rule activation visualization set it apart from conventional FNN models, making it a pioneering solution in the field of interpretable AI.

The organization of the paper is as follows: Section 2 delves into the theoretical underpinnings of fuzzy systems and the study of stress identification. Section 3 provides a detailed description of the fuzzy neural network architecture and its training methodologies. The enhancement of model interpretability through fuzzy rules is examined in Section 4. Experimental results are presented in Section 5, followed by a discussion of these findings in Section 6, culminating in the key conclusions of the study in Section 7.

2. Background

In recent years, the amalgamation of fuzzy systems with neural networks has led to the development of powerful computational tools known as fuzzy neural networks. This fusion leverages the strengths of both paradigms, offering robust modeling capabilities, particularly in domains characterized by uncertainty and imprecision. In this section, we delve into the state-of-the-art advancements in FNN, elucidating their underlying principles and applications within artificial intelligence.

2.1. Fuzzy systems and fuzzy logic neurons

Fuzzy systems, rooted in fuzzy logic, offer a mathematical framework for handling uncertainty by employing linguistic variables and fuzzy sets. Central to fuzzy logic are fuzzy sets, which generalize the classical set theory to accommodate gradual membership degrees. Fuzzy logic neurons, inspired by biological neurons, encapsulate the principles of fuzzy logic within computational units. These neurons employ fuzzy inference mechanisms to process inputs and generate output activations, effectively capturing the nuances of uncertain data [6].

2.2. Fuzzy logic operators

T-norms and t-conorms are integral to fuzzy logic, facilitating logical AND and OR operations in fuzzy systems ([18]). While t-norms identify intersections in fuzzy sets for tasks requiring strict conditions, t-conorms offer flexibility, enabling decisions based on any favorable condition. Their adaptability to n-ary and to continuous contexts is vital for developing fuzzy rule-based systems that effectively manage uncertainty and imprecision ([19]). This work primarily utilizes the conventional product (t) and probabilistic sum (s) operations to represent these concepts ([18]):

In the equations below, x and y represent the degrees of membership of elements in fuzzy sets, capturing the extent to which elements belong to the sets being operated upon.

$$t(x, y) = x \times y \tag{1}$$

The t-norm $t(x, y)$ computes the minimum degree to which elements belong to both fuzzy sets, facilitating logical AND operations.

$$s(x, y) = x + y - x \times y \tag{2}$$

The t-conorm $s(x, y)$ calculates the maximum degree to which elements belong to at least one of the fuzzy sets, supporting logical OR operations.

The uninorms [20] represent generalizations of the *t-norm* and *s-norm*, respectively, by relaxing constraints associated with a single variable, allowing it to vary within the [0,1] interval. The identity element (g) assumes values within the unit interval.

By introducing an identity element (g), uninorms extend the *t-norm* and *s-norm* by diversifying the value of g in the interval between 0 and 1. This enables a transition between an *s-norm* ($g = 0$) and a *t-norm* ($g = 1$).

It is crucial to emphasize that these binary operators (*BO*) operate in $[0, 1]^2 \leftarrow [0, 1]$ and possess monotonic, commutative, associative, and similar properties.

The specific uninorm (U) used in this study is defined as follows [21]:

$$U(x,y,g)=\begin{cases} g\cdot T\left(\frac{x}{g},\frac{y}{g}\right), & \text{if } y\le g \\ g+(1-g)\cdot S\left(\frac{x-g}{1-g},\frac{y-g}{1-g}\right), & \text{if } y>g \end{cases} \tag{3}$$

Here, T and S represent t-norm and t-conorm functions used to manage logical AND and OR operations, respectively [22]. These functions are chosen for their commutative and associative properties [23]. The parameter g dictates the balance between AND- and OR-type operations: a lower g emphasizes the OR-connection due to the lower likelihood of satisfying the first case in (3), while higher values of g facilitate more of an AND-connection. This mechanism allows for adaptive emphasis within the fuzzy logic rules, accommodating various logical operations within the same model.

2.3. Fuzzy logic neurons

Fuzzy logic neurons extend traditional neural network architectures by incorporating fuzzy inference mechanisms into their computational model. These neurons employ fuzzy rules to map input patterns into output activations, enabling nonlinear mapping in high-dimensional spaces. By leveraging fuzzy logic principles, fuzzy logic neurons excel in domains characterized by uncertainty and incomplete information, offering enhanced flexibility and adaptability [6].

The output activation y of a fuzzy logic neuron can be computed using fuzzy inference [6]:

$$y=\sum_{j=1}^{m} w_j\cdot \mu_{B_j}(x) \tag{4}$$

where w_j represents the weight associated with fuzzy rule B_j and $\mu_{B_j}(x)$ denotes the degree of activation of rule B_j given input x.

Fuzzy logic neurons leverage binary operators to aggregate fuzzy inputs and compute output activations. By incorporating binary operators within their computational model, these neurons enable the synthesis of complex fuzzy rules and the integration of heterogeneous sources of information. This flexibility in rule synthesis enhances the adaptability of FNN, enabling them to effectively model nonlinear relationships in diverse datasets.

2.3.1. And-neuron and or-neuron

Logical neurons, as described in [6], merge logical operations with learning through a fuzzy rule system, acting as nonlinear multivariable transformers within the [0,1] to $[0,1]^n$ space. These neurons process fuzzy relevance values **a** by pairing them with corresponding weights **w**, adhering to methods for logical *and* and *or* functions detailed in [6]. This integration enables the neurons to aggregate inputs and weights, showcasing the blend of logical processing and adaptability in learning within fuzzy systems. They are represented by:

$$\mathbf{z}=AND(\mathbf{w};\mathbf{a})=T_{i=1}^{n}(w_i\ s\ a_i) \tag{5}$$

$$\mathbf{z}=OR(\mathbf{w};\mathbf{a})=S_{i=1}^{n}(w_i\ t\ a_i) \tag{6}$$

where S and s are *s-norm* (typically product is used) and T and t are *t-norm* (typically probabilistic sum is used).

2.3.2. Unineuron

Utilizing uninorm principles ([20]), the unineuron combines *and* and *or* operations ([21]) in a two-tier process: L_1 for evaluating individual inputs with weights and L_2 for their collective aggregation. This approach employs t-norms and s-norms to generate outputs by first transforming $(a_i,\ w_i)$ pairs into single values $b_i = \mathbf{h}\ (a_i,\ w_i)$, then aggregating these b_i values with $\mathbf{U}\ (b_1, b_2 \ldots b_n)$, showcasing the unineuron's ability to streamline fuzzy computations through the p function ([24]).

$$p(w,a,g)=wa+\bar{w}g \tag{7}$$

using the weighted aggregation reported above the unineuron can be written as:

$$z=UNI(w;a;g)=U_{i=1}^{n}p(w_i,a_i,g) \tag{8}$$

2.3.3. Comparison between crisp and fuzzy rules

A critical distinction between crisp and fuzzy rules lies in their treatment of uncertainty. While crisp rules operate in a binary fashion, categorizing inputs as either true or false, fuzzy rules allow for gradual transitions between truth values, reflecting the inherent uncertainty present in real-world data. This nuanced handling of uncertainty endows fuzzy rule-based systems with greater expressive power and robustness, particularly in ambiguous decision-making scenarios.

For crisp rules, logical conjunction (AND) and disjunction (OR) are represented by Boolean operators:

- x and y: Inputs to the Boolean operations, typically representing truth values in binary form (i.e., 1 for true and 0 for false).

$$\text{AND}(x, y) = x \cdot y \tag{9}$$

This equation implies that both x and y must be true (1) for the result to be true (1), reflecting the nature of the Boolean AND operation.

$$\text{OR}(x, y) = x + y - x \cdot y \tag{10}$$

This represents the Boolean OR operation, where the result is true (1) if either x or y or both are true.

In contrast, for fuzzy rules, where uncertainty and partial truth values are considered, t-norms and t-conorms are utilized to compute fuzzy conjunction (AND) and disjunction (OR):

- A and B: Degrees of membership in fuzzy sets, representing truth values that range continuously between 0 and 1.

$$T(A, B) = A \cdot B \tag{11}$$

This t-norm (typically the product t-norm) computes the fuzzy AND operation, producing a result based on the product of A and B, where the result is higher only if both A and B are high, indicating a strong degree of membership.

$$S(A, B) = A + B - A \cdot B \tag{12}$$

This t-conorm (often the probabilistic sum) computes the fuzzy OR operation, ensuring that the result reflects the highest degree of membership among A and B, thus capturing the essence of OR in fuzzy logic.

2.3.4. Fuzzy neural networks and interpretability

Fuzzy neural networks integrate fuzzy logic neurons within traditional neural network architectures, offering a hybrid approach to modeling uncertain data. By combining the expressive power of fuzzy systems with the computational efficiency of neural networks, FNNs are fit for tasks requiring robustness to uncertainty and imprecision. These networks have demonstrated remarkable performance across various domains, including pattern recognition, control systems, and decision support, underscoring their utility as versatile computational tools in artificial intelligence.

In summary, the fusion of fuzzy systems with neural networks has ushered in a new era of computational modeling, characterized by enhanced robustness and flexibility in handling uncertain data. FNN stands at the forefront of this paradigm shift, offering a potent synthesis of fuzzy logic principles and neural network architectures. As research in this field continues to evolve, the potential applications of fuzzy neural networks are plentiful, promising advancements in diverse domains ranging from autonomous systems to medical diagnostics.

In 2024, the field of fuzzy neural networks (FNNs) has seen remarkable advancements across various industries, driven by significant innovations in model design and application. Bai et al. introduced a data-knowledge-driven interval type-2 fuzzy neural network that enhances interpretability and adaptability through a novel use of association rule mining techniques for rule knowledge extraction, providing a more intuitive presentation of feature relationships [25]. Zhang et al. proposed a hybrid model combining a fuzzy neural network with a variational autoencoder to improve interpretability and manage the challenge of high-dimensional unstructured data in image classification [26]. Liu et al. developed a self-organizing interval type-2 fuzzy wavelet neural network that integrates the capabilities of wavelet neural networks to improve prediction accuracy and speed, showcasing the model's effectiveness in various forecasting scenarios [27]. Lastly, Sun et al. presented an information orientation-based modular Type-2 fuzzy neural network that addresses the complexity of high-dimensional systems by improving the interpretability through a modular approach that simplifies the network topology [28].

Interpretability in fuzzy neural networks (FNNs) is essential for ensuring transparency and trustworthiness in automated decision-making processes. This subsection explores recent advancements that enhance the interpretability of FNNs, combining fuzzy logic's intuitive appeal with robust neural network learning capabilities. Emphasis is placed on methodologies that improve how FNNs manage uncertainty and imprecision in data, making them invaluable in sectors requiring clear decision-making pathways, such as healthcare, finance, and autonomous systems [29].

Distinguishability refers to the ability of a model to differentiate between inputs to generate specific outputs. In FNNs, enhancing distinguishability involves refining fuzzy rules and membership functions to better separate distinct classes or predictions. For instance, de Campos Souza et al. [30] have shown how analyzing rule distinguishability can lead to more interpretable and effective FNN architectures [30]. An illustrative diagram of distinguishable versus indistinguishable clusters is shown in Fig. 1.

Similarity measures how closely related the premises of fuzzy rules are within an FNN, affecting rule pruning and adjustment for model simplification. Techniques to measure and utilize rule similarity include adjusting the ANFIS architecture to prune redundant or closely similar rules, and enhancing model efficiency as demonstrated by Jim et al. (2024) [31]. Examples of similarity criteria used in rule comparison are depicted in Fig. 2.

Completeness in interpretability ensures that every aspect of the decision-making process is covered by the model's explanations. This is often achieved by developing exhaustive rule sets that comprehensively represent the logic underpinning the model's decisions. Pratama et al. [32] introduced a method that applies the ϵ-completeness criteria to select optimal rules for FNNs, aiming for full coverage and meaningful output generation across all input values [32]. Fig. 3 illustrates an example of evaluating model completeness.

Feature Evaluation is crucial for identifying how significant each input feature is to the model's output. This aspect of interpretability involves techniques like sensitivity analysis and feature importance ranking, which help pinpoint the contributions of

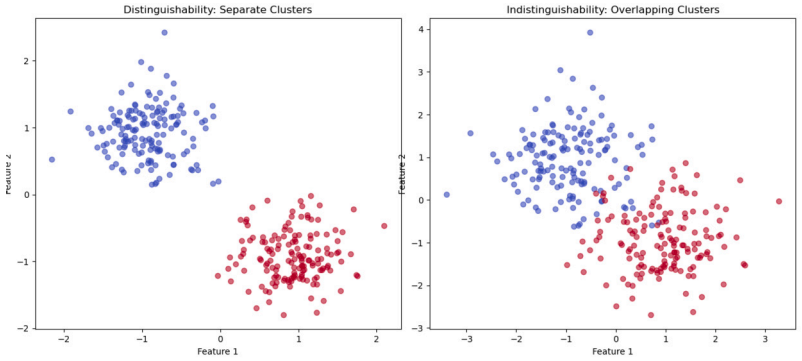


Fig. 1. Distinguishable and indistinguishable clusters.

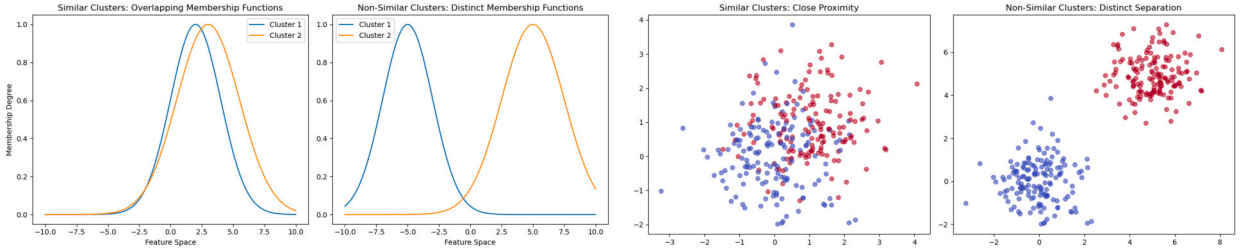


Fig. 2. Similarity criteria.

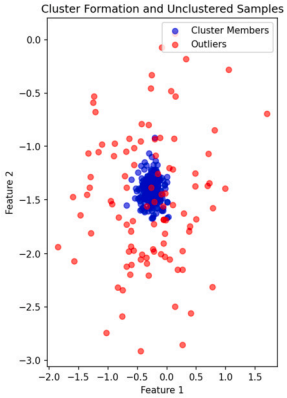


Fig. 3. Completeness evaluation.

individual features to decision processes. Liu et al. [33] introduced an FNN architecture that incorporates mechanisms for direct interpretation of feature impacts, enhancing transparency [33] (see Fig. 4).

Interpretation of Consequents focuses on the output layer of FNNs, where decisions are finalized. Enhancing this aspect involves simplifying and clarifying the rule consequents to make output decisions more transparent to users. De Campos Souza and Lughofer [34] explored methods to achieve clearer interpretation of consequents within FNNs, promoting easier understanding of model outputs [34].

Collectively, these advancements in FNN interpretability not only improve the methodologies for developing these networks but also refine the techniques for assessing their clarity and effectiveness. As FNN technology progresses, maintaining a strong focus on interpretability ensures their suitability for critical decision-making applications, fostering greater trust and transparency in automated systems.

Recent strides in FNN have merged deep learning and fuzzy logic, enhancing both accuracy and transparency across diverse sectors. Hassani et al. [35] focused on data fusion in structural monitoring and enhancing model generalization in classifiers, pointing to the crucial role of interpretability [35]. Liao et al. [36] and de Campos Souza and Lughofer [37] explored machine learning in decision-making and evolving fuzzy classifiers, further advocating for models that integrate expert rules and manage uncertainties with clarity [36,37].

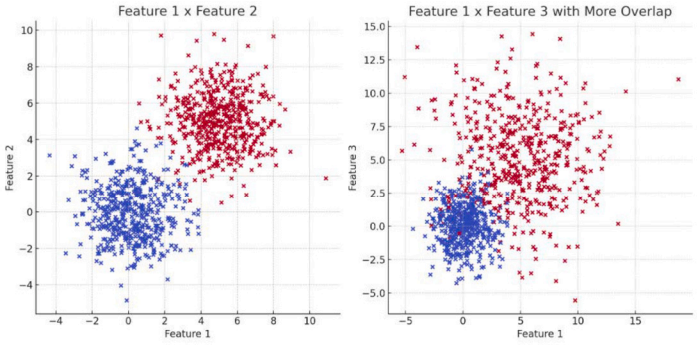


Fig. 4. Feature evaluation.

3. IFNN: architecture, training, and interpretable tools

The IFNN framework introduced in this study represents a significant advancement in fuzzy neural networks (FNNs), aiming to enhance both interpretability and performance optimization by integrating the Adam optimization algorithm. Building on the foundational principles set by Campos Souza (2008) [17], which combined fuzzy logic with neural network techniques for pattern recognition, this framework focuses on optimizing performance while improving interpretability.

Key to the IFNN is its sophisticated fuzzification process, using Gaussian neurons and uniformly distributed membership functions to translate input features into the fuzzy domain. This step lays the groundwork for a comprehensive fuzzy inference system designed to decode complex data relationships.

The framework further introduces an intermediate layer of 'and' or 'or' logic neurons, enhancing its pattern recognition capability by building on Campos Souza's conceptual base [17] with modern improvements for current challenges in interpretability.

A major innovation is the adoption of the Adam optimization algorithm during training, optimizing rule updates and retaining the full set of fuzzy rules for detailed post-training analysis. This approach not only facilitates an in-depth examination of data and insightful visualizations but also maintains the entirety of learned fuzzy rules, significantly enriching the model's interpretability and analytical depth.

In essence, the IFNN framework marks a pivotal step in combining fuzzy logic with neural networking, highlighted by the strategic use of the Adam optimization algorithm to achieve a balance between performance optimization and interpretability.

3.1. *Fuzzy neural network architecture: structure of variable neurons and activation functions*

The IFNN framework distinguishes itself by embedding interpretable patterns within its outcomes and emphasizes the importance of interpretability monitoring. Central to this framework is the novel application of a grid partition fuzzification process at its initial layer, comprising L neurons based on equally spaced membership functions—a departure from traditional density-derived functions. This process allows for the establishment of L membership clouds (A_{lj}) for each input feature x_{ij}, where the membership functions serve as activation functions. This setup initiates fuzzification, producing membership degrees (a) linked to input values, represented as $a_{jl} = \mu_l^A$ for $j = 1...\ N$ and $l = 1 \ldots L$. Importantly, the number of neurons L in this layer is determined by the power relation $L = N^N$, based on the number of fuzzy sets per input (M) and the dimensionality of the problem, reflecting the combinatorial increase in complexity with more dimensions and membership functions. This grid-based approach ensures a systematic and comprehensive fuzzification, crucial for the framework's optimized performance and enhanced interpretability.

The model's rule-building layer features L logical neurons (andneurons- Eq. (5), orneurons- Eq. (6), unineurons- Eq. (8)). Each neuron is tasked with a weighted aggregation of first-layer outputs using feature weights w_{il} (for $i = 1...\ N$ and $l = 1...\ L$), assigned randomly. For each input variable j, a single output a_{jl} from the initial layer is selected as input to the l-th neuron, facilitating the derivation of fuzzy rules from the dataset.

The defuzzification process in the third layer of the IFNN model utilizes an aggregation neural network with a single neuron that applies a sigmoid activation function, defined as:

$$f_\Gamma(z) = \frac{1}{1 + e^{-z}} \tag{13}$$

where z represents the input to the neuron, which is the aggregated fuzzy value processed by the fuzzy neurons (including andneurons, orneurons, or unineurons). These neurons translate fuzzy inputs into a single value which is then transformed by f_Γ from fuzzy logic to a crisp output, critical for addressing both regression and classification tasks effectively.

The overall output of the network, Ω, is determined by comparing the sum of the outputs of f_Γ across all l neurons against zero, as illustrated below:

$$\Omega = \begin{cases} 1, & \text{if } \sum_{j=0}^{l} f_\Gamma(z_j, v_j) > 0 \\ 0, & \text{if } \sum_{j=0}^{l} f_\Gamma(z_j, v_j) \leq 0 \end{cases} \tag{14}$$

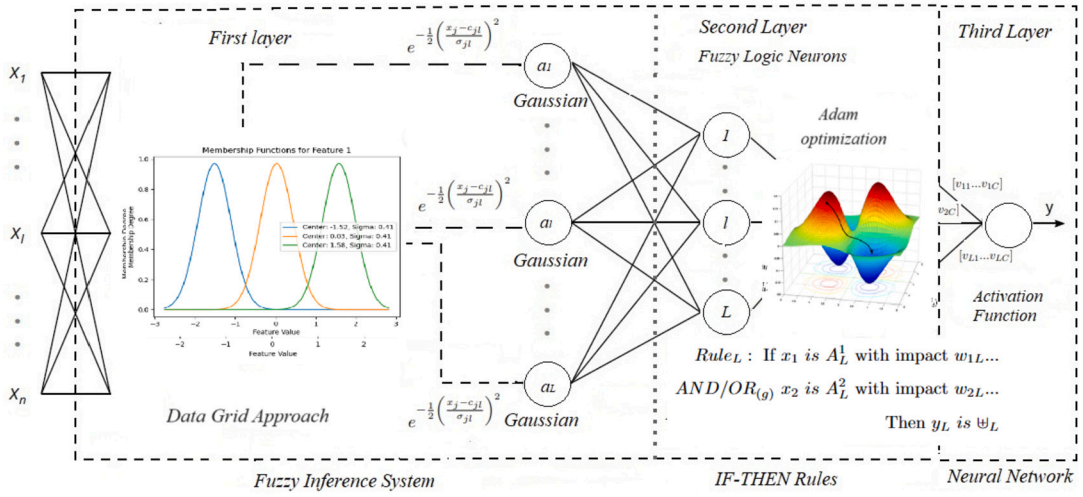


Fig. 5. FNN architecture.

Here, z_j denotes the input from the jth fuzzy neuron (e.g., andneuron, orneuron, or unineuron), and v_j is the corresponding weight for that neuron, dynamically adjusted by the Adam optimizer to optimize performance.

This structured approach, inspired by the work of Campos Souza [38] and further refined with the Adam optimization algorithm, positions the IFNN as a significant advancement in the realm of FNN. The innovative application of the Sigmoid activation function, coupled with a methodical approach to fuzzification and rule construction, exemplifies the framework's capability to enhance interpretability and optimization in complex pattern recognition endeavors.

Fuzzy rules can be extracted from the network topology in the form as presented in Eq. (15) below:

$$\begin{aligned} Rule_1:\ &\text{If } x_1 \text{ is } A_1^1 \text{ with impact } w_{11}\ldots \\ AND/OR_{(g)}\ &x_2 \text{ is } A_1^2 \text{ with impact } w_{21}\ldots \\ &\text{Then } y_1 \text{ is } \vec{v}_1 \\ Rule_L:\ &\text{If } x_1 \text{ is } A_L^1 \text{ with impact } w_{1L}\ldots \\ AND/OR_{(g)}\ &x_2 \text{ is } A_L^2 \text{ with impact } w_{2L}\ldots \\ &\text{Then } y_L \text{ is } \vec{v}_L \end{aligned} \tag{15}$$

where

- x_1, x_2: Input variables to the fuzzy system.
- $A_1^1, A_1^2, A_L^1, A_L^2$: Fuzzy sets associated with each input variable in the respective rules, denoted by A, which are the output of Gaussian functions from the first layer of the network. Each fuzzy set represents a specific linguistic label like "low," "medium," or "high."
- $w_{11}, w_{21}, w_{1L}, w_{2L}$: Weights or impact factors for each fuzzy set, quantifying the influence of the respective input variable on the rule's output.
- g: A parameter from uninorm concepts that determine the use of logical connectives between fuzzy terms within a rule. Depending on the value of g, the logical connective may behave more like an AND or an OR, allowing for a flexible interpretation of the input data.
- $\vec{v}_1, \ldots, \vec{v}_L$: Consequent vectors representing the output of each rule, specifically tailored for binary classification tasks. These consequents are usually singletons, each corresponding to a particular class or decision outcome.

The logical operations "AND/OR_(g)" dynamically adjust between AND-like and OR-like behaviors based on the parameter g, enabling the model to adapt the decision-making process to varying levels of input certainty and rule specificity.

Fig. 5 illustrates the FNN's feed-forward structure, showcasing the fuzzification process, the generation of fuzzy neurons in the second layer, and the third layer's role in aggregation. This setup forms a neuro-fuzzy inference system that adeptly generates fuzzy rules, merging neural and fuzzy computation to interpret complex data patterns. The final layer acts in defuzzification, transforming fuzzy inferences into precise outputs. This enhances the system's interpretability and data classification capabilities.

The weights of the output neural network, refined through optimization via the Adam technique, capture the conclusions of the fuzzy inference system. Consequently, the rules extracted by the first two layers articulate the significance of integrating various features of the problem in relation to the anticipated solutions. This process underscores the critical role of feature amalgamation in delineating the contours of expected outcomes, facilitated by the interplay of neural network optimization and fuzzy logic inference.

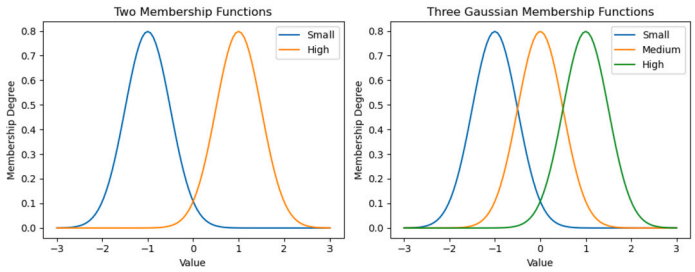


Fig. 6. Diverse interpretations generated from different partitions.

3.2. First layer - grid-based fuzzification with uniform membership functions

The first layer of the FNN model is distinguished by its grid-based fuzzification methodology, a critical strategy for converting raw input data into an organized ensemble of fuzzy sets. This approach is essential for achieving a uniform distribution of membership functions across the spectrum of input variables, thereby facilitating coherent fuzzy rule generation [39].

In the model, each input variable x_j is linked to multiple fuzzy sets A_{lj}, which act as activation functions within the fuzzy neural network. These relationships are crucial for determining how well each input matches the conditions described by the fuzzy sets. The relationship is expressed mathematically as:

$$a_{jl} = \mu_l^A(x_j), \quad \text{for } l = 1, \ldots, M. \tag{16}$$

Where:

- a_{jl} represents the membership value of the input x_j in the fuzzy set A_{lj}.
- μ_l^A denotes the membership function for the l-th fuzzy set, applied to the j-th input.
- M is the total number of fuzzy sets associated with each input, reflecting the model's capacity to capture varying degrees of membership across its input range.

The fuzzy sets A_{lj} are designed using Gaussian membership functions, strategically arranged with similar support to comprehensively cover the domain. Each membership function is tailored to attain a height of 1 (normality), ensuring that the union of all membership degrees fully represents the domain. Moreover, a delta coverage of 0.5 is maintained across the domain to ensure a precise and accurate depiction of the problem domain. This systematic coverage allows the model to finely tune its response to different regions of the input space, enhancing its ability to handle diverse data distributions and making the model particularly suited for applications requiring detailed analysis of input variations.

The degree of membership for an input x_j within a fuzzy set, calculated by the Gaussian neuron's activation function, is defined as:

$$\omega(x_j, c_{jl}, \sigma_{jl}) = e^{-\frac{1}{2}\left(\frac{x_j - c_{jl}}{\sigma_{jl}}\right)^2}, \quad \text{for } j = 1, \ldots, N, \quad l = 1, \ldots, M, \tag{17}$$

where N represents the number of input variables, M indicates the number of fuzzy sets per input and c_{jl} and σ_{jl} denote the center and standard deviation of the Gaussian function for the lth fuzzy set associated with the jth input variable, respectively.

Additionally, the weights for the Gaussian neurons, w_{il}, are initialized within the range [0,1] to modulate the influence of each neuron's output:

$$w_{il} \text{ for } i = 1, \ldots, N, \text{ and } l = 1, \ldots, M. \tag{18}$$

This initialization strategy not only determines how the inputs are fuzzified into respective membership degrees but also how these fuzzy inputs interact within the network to form the basis for decision-making processes. The precise articulation of these parameters and the systematic layout of the fuzzification process ensure that the input data are effectively transformed into meaningful fuzzy sets. These sets form the foundational components for generating and interpreting fuzzy rules within the FNN model, thereby facilitating accurate and reliable fuzzy logic-based reasoning and decision-making.

The adoption of equally spaced membership functions greatly enhances the visualization clarity of analyzed patterns, as evidenced in Fig. 6, which presents how varying the number of partitions can change the result of the problem interpretation. The challenges of term delineation, particularly when overlaps are encountered as shown in Fig. 2, represent different divisions and possible interpretations of the fuzzy sets. For example, in the first case, the MF can be a representation of cold and hot, and in the second case, the inclusion of another MF changes the interpretation of this MFS.

Ultimately, the grid data partitioning outlines the decision space of the neurons, demonstrated in Fig. 7, which provides an example of a grid partition with two membership functions. This illustrates the nuanced approach to model partitioning, highlighting the balance between accuracy and interpretability, while adhering to cognitive limitations on dimensionality to foster intuitive and transparent decision-making frameworks in the realm of FNNs.

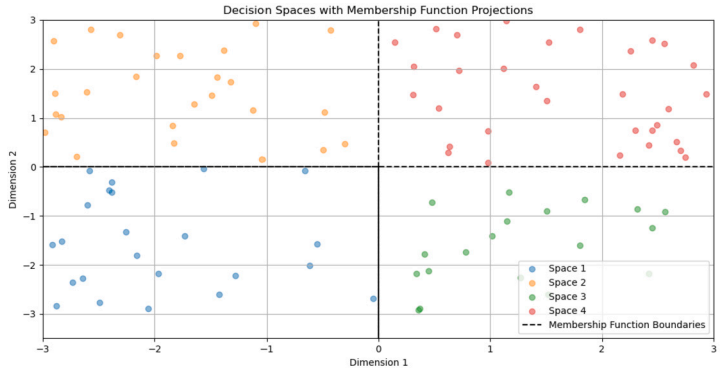


Fig. 7. Example of a grid partition with 2 membership functions, illustrating the decision space construction.

FNN utilizes grid-based fuzzification and therefore introduces a degree of complexity, particularly through the interplay between the number of features x and membership functions x per feature. This leads to an exponential neuron increase, depicted as $L = M^N$. This relationship between feature count and model complexity, while potentially enhancing accuracy, may compromise interpretability—a critical aspect of fuzzy system design.

Echoing insights from cognitive science, particularly Miller's (1956) [40] on human cognitive limitations, suggests an interpretability threshold within a seven-dimension framework. This cognitive boundary underscores the importance of balancing model complexity with human interpretability, advocating for a measured approach to feature selection and partitioning to maintain model usability within human cognitive limits.

Thus, designing FNNs with grid-based fuzzification requires careful consideration of feature selection and model partitioning to ensure a balance between precision and interpretability, in line with cognitive constraints. This balance not only makes FNNs more user-friendly but also supports the goal of creating intuitive and transparent decision-making tools.

3.3. Third layer - neural aggregation output layer

The third layer of the Fuzzy Neural Network (FNN) architecture executes the defuzzification process, where it aggregates outputs from the second layer's fuzzy neurons, transitioning from fuzzy logic interpretations to a quantifiable, crisp output. This crucial transition is defined mathematically by the following sophisticated aggregation formula:

$$\hat{y} = \Omega\left(\Gamma\left(\upsilon_0 + \sum_{j=1}^{L} z_j \cdot \upsilon_j\right)\right), \tag{19}$$

where Γ represents the Sigmoid activation function (referenced in Eq. (13)), mapping input values to a bounded interval (0,1), making it ideal for binary classification tasks and interpreting fuzzy logic rules, Ω (described in Eq. (14)). The element confines the aggregated neural inputs to ensure outputs remain within the expected range, further enhancing decision-making clarity. υ_0 acts as a bias term, with $z_0 = 1$ establishing a constant offset that adjusts the activation threshold. This adjustment allows for more flexible decision boundaries, accommodating diverse data distributions. z_j denotes the outputs from the second layer's fuzzy neurons. υ_j for $j = 1, \ldots, L$ are the corresponding weights, integral to the integration and mapping of these outputs through the Sigmoid function to display a probability space, culminating in the model's predictive output $\hat{y}$.

This configuration not only ensures the effective conversion of fuzzy interpretations into actionable outputs but also highlights the network's capability to manage and interpret complex fuzzy rules, facilitating robust decision-making processes within the FNN architecture.

This third layer, by employing the Sigmoid function, transitions fuzzy inferences to precise outputs, demonstrating the network's capacity to interpret fuzzy logic probabilistically. This process not only finalizes the defuzzification but also ensures the model's effectiveness in producing actionable insights from complex data patterns, highlighting the strategic design of the FNN's aggregation layer for optimal interpretability and application.

Moreover, the strategic placement of these examples spans across the Sigmoid curve, encapsulating scenarios with negative, neutral, and positive aggregated outputs. This diversity not only underscores the Sigmoid function's capacity to handle a wide array of input values but also its critical role in ensuring that the defuzzification process yields outputs that are inherently probabilistic, thus facilitating decision-making processes that are grounded in statistical certainty.

3.4. Training of the third layer with Adam optimizer

The training of the model's third layer utilizes the Adam optimizer, renowned for its robustness and adaptability, particularly in optimizing neural network parameters. The Adam optimizer amalgamates the strengths of both the Adaptive Gradient Algorithm (AdaGrad) and Root Mean Square Propagation (RMSProp), thereby adeptly managing sparse gradients in noisy environments.

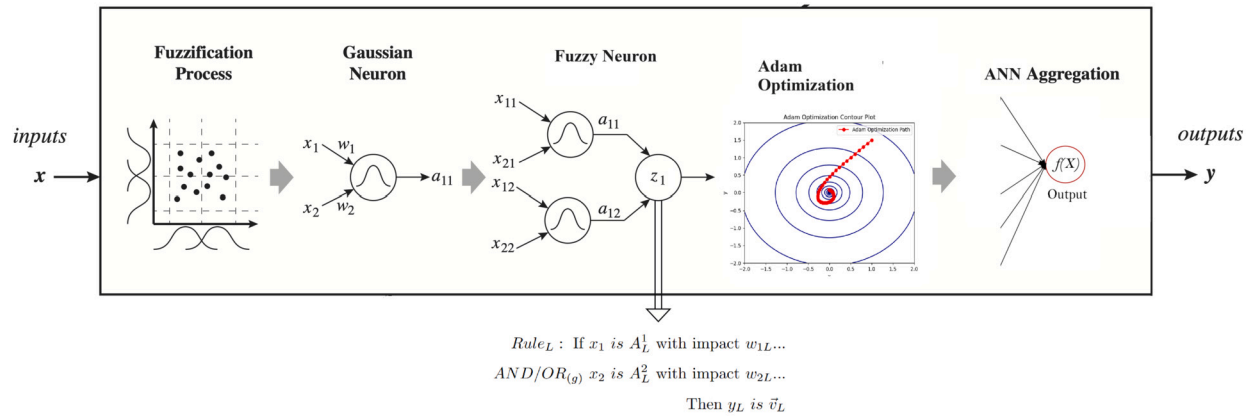


Fig. 8. IFNN work flow.

In the IFNN model framework, the Adam optimizer plays a critical role in refining the weights of neurons, including andneurons, orneurons, and unineurons, as detailed in Eq. (5), Eq. (6), and Eq. (8). It accomplishes this through iterative adjustments that rely on adaptive estimations of the gradients' first ($\hat{m}_t$) and second moments ($\hat{v}_t$), enhancing the model's learning dynamics. The formula for updating the weights via the Adam optimizer is provided below:

$$\vec{v}_k^{(t+1)} = \vec{v}_k^{(t)} - \frac{\eta \cdot \hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}, \tag{20}$$

where:

- η represents the learning rate, essential for controlling the step size during weight updates.
- $\hat{m}_t$ and $\hat{v}_t$ are the biased-corrected first and second-moment estimates of the gradients, crucial for stabilizing and expediting the convergence process.
- ϵ is a small constant added to prevent division by zero, enhancing the numerical stability of the optimizer.

The implementation of this adaptive weight adjustment technique underlines the robustness of the Adam optimizer, making it exceptionally suited for handling diverse computational challenges in neural network training.

Within the IFNN model, the training culminates in the precise optimization of the last layer's weights, which is pivotal for the accurate construction and interpretation of fuzzy rules. The final weights, denoted by $\vec{v}_k$, are computed using the Adam optimizer as follows:

$$\vec{v}_k = \text{Adam}(\vec{Z}, \vec{y}_k) \quad \forall k = 1, \ldots, C, \tag{21}$$

where C indicates the number of classes, reflecting the model's capacity to handle multi-class scenarios. $\vec{Z}$ encapsulates the activation outputs of all neurons across input samples, integrating the individual neuron responses and $\vec{y}_k$ corresponds to the target output vector for each class, tailored specifically to match the sigmoid function's output, facilitating precise binary classification.

This refinement of $\vec{v}_k$ through the Adam optimizer significantly enhances the model's precision and interpretability in outcome prediction, demonstrating the profound impact of optimized weight adjustments on the performance of fuzzy neural networks.

The only parameter of the IFNN model is the grid partition parameter (M), which governs the fuzzification approach. The training algorithm, encapsulating the entire process from initialization through optimization, is outlined in Algorithm 1 and Fig. 8:

Algorithm 1 Fuzzy Neural Network Training Process with Adam Optimization.

Input: Data set $\vec{X}$, target vector $\vec{y}$.
Step 1: Initialize grid partition using membership functions (M) for each input feature.
Step 2: Calculate centers (c_{jl}) and standard deviations (σ_{jl}) for L fuzzy sets per input feature, ensuring uniform distribution.
Step 3: Construct L fuzzy neurons per feature using Gaussian membership functions, as per Eq. (16).
Step 4: Generate fuzzy neuron outputs ($\vec{Z}$) for each input sample.
Step 5: Adapt $\vec{y}$ for binary classification.
Step 6: Employ the Adam optimizer to train the neural network, guided by Eq. (21).
Step 7: Update weight parameters $\vec{v}$ to reflect the optimized weights from the neural network's final layer.
Output: Trained IFNN with optimized fuzzy rules.

4. Advanced interpretation of the rules extracted from the fuzzy neurons

Interpretable model architectures and operations help model users understand and accept computational techniques In this section, methods for evaluating the interpretability of the IFNN are presented.

4.1. Auxiliary methods for interpretability

The quest for interpretability within FNNs is paramount, particularly in domains where decision-making impacts are significant, such as healthcare and finance. Enhanced interpretability ensures that the mechanisms behind AI decisions are transparent, fostering trust in automated systems. We explore three advanced techniques aimed at elucidating the logic and decision-making processes of FNNs.

4.2. Consistency in FNN

In interpretable machine learning, especially in fuzzy system modeling, consistency is crucial for the reliability and interpretability of models. It ensures logical coherence within a fuzzy rule base, requiring that similar premises lead to similar consequents to prevent contradictory rules that could impair the system's decision-making.

Considering a fuzzy rule base R from fuzzy rule-based systems (FRBS) with L rules, each rule R_i comprises a premise P_i and a consequent C_i. The consistency between any two rules, R_i and R_j, depends on the similarity of their premises P_i and P_j, and their consequents C_i and C_j. Premise similarity can be measured by metrics like Euclidean distance, while consequent similarity may be assessed through linguistic term closeness or numerical closeness. The mathematical expression for rule consistency between R_i and R_j is:

$$\text{Consistency}(R_i, R_j) = \exp\left(-\sqrt{\Delta P_{ij}^2 + \Delta C_{ij}^2}\right) \tag{22}$$

where ΔP_{ij} denotes the difference or distance between the premises of R_i and R_j, and ΔC_{ij} signifies the difference between the consequents of R_i and R_j. The exponential function here is utilized to transform the aggregate differences into a similarity score that ranges between 0 and 1.

In this paper, consistency is quantified by ΔP_{ij} and ΔC_{ij}, measuring the distance between premises and consequents of rules R_i and R_j. Using an exponential function, these distances are transformed into a similarity score ranging from 0 to 1, where scores near 1 indicate strong logical coherence, and scores closer to 0 highlight discrepancies needing correction. Achieving high consistency requires designing fuzzy sets and rules to minimize premise overlap with inconsistent consequents, potentially involving rule-based pruning to improve decision-making clarity and system interpretability.

4.3. Similarity of fuzzy neurons

In interpretable fuzzy systems, assessing the similarity of rules is pivotal for refining the rule base to improve interpretability and efficiency. This assessment involves the use of a similarity matrix, denoted by $S \in \mathbb{R}^{L\times L}$, where L represents the number of rules. The matrix is used to compare the premises of fuzzy rules by evaluating the resemblance between their centers (c) and standard deviations (σ), which are critical parameters for defining the antecedents of the rules.

The similarity matrix is computed using the following formula:

$$s_{ij} = \exp\left(-\sqrt{\sum_{k=1}^{D}(c_{ik} - c_{jk})^2 + (\sigma_{ik} - \sigma_{jk})^2}\right), \tag{23}$$

where:

- D represents the dimensionality of the input space, accounting for the number of input variables each rule considers.
- c_{ik} and c_{jk} denote the centers of the ith and j^{th} rule's fuzzy sets in the kth dimension, respectively.
- σ_{ik} and σ_{jk} indicate the standard deviations of the fuzzy sets for the ith and j^{th} rules in the kth dimension, respectively.

This similarity measure, employing the Euclidean distance between rule parameters followed by an exponential function, transforms the calculated distances into scores within the range of $[0, 1]$. High similarity scores, approaching 1, suggest that the rules are very similar, whereas low scores, near 0, indicate significant differences. This method facilitates the identification of redundant rules or areas needing additional coverage, enabling rule set optimization by pinpointing opportunities for pruning or consolidation to enhance the model's interpretability without scarifying much of its performance.

A similarity matrix, consisting of symmetric and sized $L \times L$ for L rules, evaluates rule redundancy by measuring the similarity between any two rules, with self-similarity marked as 1 on diagonal elements. This matrix aids in identifying redundant rules, and guiding optimizations to improve the fuzzy system's interpretability and efficiency. Streamlining the rule base through merging or adjusting similar rules enhances decision-making clarity and reliability.

4.4. Evaluation of e-completeness in Fuzzy Rule-Based Systems

The interpretability of fuzzy rule-based systems (FRBS) significantly benefits from the metric of ϵ-completeness, which quantifies the system's capability to yield meaningful outputs across all possible inputs. This is predicated on the criterion that for any given input, at least one rule's activation level must exceed a predefined threshold, ϵ, thus ensuring that the system is comprehensive in its coverage.

Mathematically, the ϵ-completeness for an FRBS with a set of rules $R = \{r_1, r_2, ..., r_n\}$ over an input space X is defined as follows:

$$\epsilon\text{-Completeness}(X, R, \epsilon) = \frac{1}{|X|} \sum_{xinX} \mathbb{1} \left(\max_{iin\{1,...,n\}} \mu_{r_i}(x) \geq \epsilon \right), \tag{24}$$

where $\mu_{r_i}(x)$ represents the activation of the ith rule for input x, and $\mathbb{1}$ is the indicator function that equals 1 if the condition within the parenthesis is met, and 0 otherwise. This formula ensures that the system is tested for reliability across the entirety of X, highlighting its ability to consistently activate at least one rule above the threshold ϵ, thereby substantiating its interpretability and reliability. The threshold ϵ is typically set based on application-specific requirements, balancing sensitivity and specificity to ensure that the FRBS adequately captures the nuances of the input data while avoiding overfitting.

Our revised methodology for fuzzy rule-based systems (FRBS) enhances rule activation calculations by utilizing outputs from the logical layer (Z), making the analysis more efficient. This approach allows for direct sourcing of activation levels $\mu_{r_i}(x)$ from Z, eliminating the need for repetitive rule evaluations. To assess the system's coverage and identify areas needing refinement, we introduce a visualization technique that maps rule activations against test inputs, with color intensity indicating activation levels. This visualization, combined with ϵ-completeness, provides a comprehensive assessment of the FRBS's coverage, highlighting areas for targeted improvements to enhance the system's adaptability and effectiveness. This methodology underscores the importance of thorough rule coverage and visual analysis in optimizing FRBS performance.

4.5. Distinguishability in FNN

Distinguishability is a relevant concept in fuzzy systems, particularly in fuzzy neural networks, where it ensures that each fuzzy set (or membership function, MF) distinctly represents linguistic terms across the input space. This distinctness is crucial for the interpretability and effectiveness of fuzzy rule-based models, facilitating the assignment of clear semantic meanings to the fuzzy sets and ensuring that the model's decisions can be understood and justified.

The distinguishability of a set of fuzzy membership functions in an FNN can be quantitatively assessed through a measure that evaluates the overlap between pairs of membership functions. For a given input feature, let us consider a set of membership functions $\{MF_1, MF_2, ..., MF_n\}$, where each MF_i is defined by its center c_i and width (standard deviation) σ_i. The distinguishability between any two membership functions, MF_i and MF_j, can be calculated as follows:

$$D(MF_i, MF_j) = \exp\left(-\sqrt{(c_i - c_j)^2 + (\sigma_i - \sigma_j)^2} \right), \tag{25}$$

where $D(MF_i, MF_j)$ is the distinguishability measure between MF_i and MF_j, ranging from 0 (completely indistinguishable) to 1 (completely distinguishable).

4.6. Computational complexity analysis with parameter nomenclature

The computational complexity of the IFNN algorithm involves several critical stages, each characterized by specific operations and computational demands. By referencing the parameters detailed in previous sections, we elucidate the computational complexity of each stage within the IFNN model's lifecycle.

Fuzzification Layer Complexity: The fuzzification layer transforms the input vector $\vec{x}$ into fuzzy set activation levels $\vec{a}$ using Gaussian membership functions characterized by centers $\vec{c}_i$ and standard deviations $\vec{\sigma}_i$. Given N inputs (features) and k membership functions per feature, the complexity of this layer is $O(n \cdot N \cdot k)$, where n represents the number of samples.

Logic Neurons Layer Complexity: After fuzzification, the logic neurons layer processes the fuzzy outputs through a combination of t-norms $t(x, y)$ and t-conorms $s(x, y)$, leading to a total of $L = k^N$ extracted neurons. This layer exhibits a complexity of $O(n \cdot k^N)$.

Training with Adam Optimizer Complexity: During training, the Adam optimizer updates weight vectors $\vec{v}_k$ and bias terms b based on gradients $\nabla_\theta J(\theta)$ of the loss function $J(\theta)$. Assuming e epochs, a batch size of b, and a computational cost q for processing one batch, the training complexity is $O(e \cdot \frac{n}{b} \cdot q)$.

Model Evaluation Complexity: Evaluating the model involves calculating metrics based on the predicted output $\hat{y}$ and the real class labels y_k. For n_{test} test samples, the evaluation complexity is $O(n_{test})$.

Interpretability Evaluation Complexity: The interpretability evaluation, involving the calculation of similarity S and consistency matrices, as well as the e-completeness measure ϵ-Completeness(X, R, ϵ), adds another layer of complexity. Let r denote the number of logic neurons, resulting in a complexity of $O(r^2)$ for these interpretability metrics.

Overall Computational Complexity: Combining all components, the final equation for the computational complexity of the FNN model, from data preprocessing through to training, evaluation, and interpretability analysis, is given by:

$$O(n \cdot N \cdot k) + O(n \cdot k^N) + O(e \cdot \frac{n}{b} \cdot q) + O(n_{test}) + O(r^2) \tag{26}$$

Table 1
Dataset Dimensions and Training/Test Split.

Dataset	Instances	Attributes	Training Set	Test Set
Iris Dataset	150	4	105	45
Mammographic Masses	961	5	673	288
Haberman's Survival	306	3	214	92
Blood Transfusion	748	4	524	224
Liver Disorders	345	6	242	103

This equation highlights the comprehensive computational demands of deploying an IFNN model, showcasing the balance required between model complexity, interpretability, and computational efficiency for practical applications.

5. Experimental evaluation

This section delineates the experimental framework designed to substantiate the efficacy of the proposed model in elucidating interpretability within the context of binary problem-solving scenarios. The empirical investigation is partitioned into three distinct subsections, each aimed at illuminating various facets of the model's operational dynamics and its interpretive output.

5.1. Interpretability insights through synthetic data exploration

This initial subsection is devoted to an in-depth exploration of the interpretability capabilities of our model, facilitated by a meticulously engineered synthetic dataset. The primary objective of this exploration is twofold: firstly, to demonstrate the model's proficiency in navigating the complexities of binary classification problems, and secondly, to unravel the nuanced concepts that underpin both the utilization of the model and the interpretation of its outcomes.

The adoption of a synthetic dataset in this context serves as a pivotal tool, creating a controlled environment that is conducive to a detailed examination of the model's operational dynamics. Such an environment not only permits an unobstructed view of the model's decision-making processes but also enables a systematic dissection of the model's interpretative framework. This framework, crucial for understanding the model's internal logic, hinges on several key interpretability techniques, including the visualization of logic neuron outputs, the projection of these outputs in a three-dimensional space, and the computation of similarity matrices for fuzzy rules.

In our analysis, we iterate over various configurations of the model, each distinguished by the type of neuron (andneuron, orneuron, unineuron) and the number of membership functions. This iterative process is instrumental in quantifying the impact of these configurations on the model's accuracy and

5.2. Comparative analysis with state-of-the-art models

This section presents a comparative analysis of the IFNN model against established models in binary pattern classification, highlighting its performance, accuracy, and notably, its interpretability. The study evaluates IFNN alongside traditional and ensemble learning methods like the Multilayer Perceptron (MLP), Naive Bayes, and Random Forest, using a diverse set of datasets (Table 1) including Iris Dataset, Mammographic Masses, Haberman's Survival, Blood Transfusion Service Center Dataset, and Liver Disorders, reflecting a diverse array of data types and sources, to assess the model's classification capabilities and robustness. Central to the model's evaluation is its tri-layered architecture that enhances interpretability by converting logical neurons into explicit fuzzy rules. The effectiveness of IFNN is quantified using metrics such as Accuracy, Precision, Recall, and F1 Score, demonstrating its ability to offer clear insights into its decision-making process. These metrics collectively provide a nuanced insight into the efficacy of the model in classifying the datasets under study.

- **IFNN Models:** The variants IFNN AndNeuron, IFNN OrNeuron, and IFNN UniNeuron were scrutinized, with the Adam optimization algorithm serving as a pivotal enhancement to their configuration with 4 membership functions for each dimension.[1]
- **Benchmark Models:** The Multilayer Perceptron [41] was endowed with a single hidden layer of 100 neurons to ensure sufficient complexity, while the Random Forest Classifier [42] was equipped with 100 decision trees, augmenting the model's accuracy and resistance to overfitting. The Gaussian Naive Bayes model [43] was employed in its default setup, offering a probabilistic perspective for comparison.

This study employs a 70:30 split for training and testing the models, adhering to widely recognized machine learning practices. The intent is to verify whether the IFNN model not only competes with but possibly exceeds the performance of conventional models across accuracy, precision, recall, and F1 score metrics.

In this study, we extend our evaluation of the IFNN model beyond comparative performance analysis by incorporating a rigorous statistical assessment. Our methodology is grounded in conducting 30 repetitions of the experiment, a strategy that solidifies the

[1] After previous experiments with mf = [2,3,4,5,].

reliability and robustness of our results. This thorough approach allows for an accurate comparison of the IFNN model's performance against established benchmarks.

For statistical analysis, we utilize the Analysis of Variance (ANOVA) [44] to examine the accuracy metrics obtained from these repetitions, identifying any significant performance differences among the models tested. In cases of significant findings from ANOVA, we proceed with post-hoc tests, such as Tukey's Honestly Significant Difference (HSD) [45], to determine the specific models among which these differences lie. Additionally, we ensure the validity of our statistical approach by testing for normality with the Shapiro-Wilk test [46] and assessing variance homogeneity through Levene's test [47].

5.3. Application to a real-world academic dataset: sepsis dataset

In the final segment of our experimental validation with IFNN, we delve into the application of our proposed models to a real-world dataset, which originates from seminal research on sepsis. This exploration is not merely a demonstration of the model's capability to adapt and perform in real-world conditions but also a testament to its ability to derive meaningful, interpretable insights from complex data landscapes. By applying our models to empirically sourced data, we underline the invaluable role of such models in both academic research and practical healthcare applications, where the stakes of interpretability and model transparency couldn't be higher.

To adapt and extend the significance of AI in sepsis identification real-world datasets, the integration of advanced AI algorithms showcases the potential to provide early detection and classification of sepsis. The emergence of sophisticated models, as evidenced in recent studies, illuminates the path toward significantly enhancing patient outcomes through early and accurate sepsis prediction.

5.3.1. Experimental setup

Our experimental framework evaluates the performance and knowledge extraction capabilities of IFNN models in three configurations—AndNeuron, OrNeuron, and UniNeuron variants, all optimized by Adam algorithm and equipped with four membership functions, and benchmark models including RandomForestClassifier [42], MLPClassifier [41], GaussianNB [48], SVC [49], and LogisticRegression [50], all split in a 70/30 train-test ratio.

This systematic comparison aims to not only highlight the efficiency of our models with respect to the state-of-the-art but also to meticulously dissect the knowledge extracted by the best-performing IFNN model. Such insights are pivotal for clinical decision-making and advancing sepsis research.

Moreover, we incorporate a detailed evaluation of the three IFNN model variants, focusing on their unique contributions to performance and interpretability. The evaluation metrics remain consistent with those outlined in the previous section, ensuring a coherent and comparative analysis.

The dataset characteristics and the distribution of sepsis cases are thoroughly outlined for clarity. Key variables include the patient's age in years, which lacks missing values and is measured in years, providing insight into the age distribution among the patients. Gender is represented as 'sex' with '0' indicating male and '1' indicating female, and there are no missing entries for this variable. The 'episode number' variable tracks the number of prior sepsis episodes experienced by a patient, again without missing data, offering a perspective on the recurrence of sepsis within the dataset. Lastly, the 'hospital outcome' variable records the patient's status after 9,351 days as either '1' for Alive or '0' for Dead, with no missing values, providing a crucial endpoint for analysis. This comprehensive framework supports a detailed investigation into sepsis outcomes based on patient demographics and clinical history.

6. Discussion about the tests

This section aims to analyze experimental results to validate initial hypotheses and contribute new insights to the field, enhancing both current understanding and future research.

6.1. Analysis of performance metrics

The exploration of our model's performance across varying configurations (according to section 5.1) provides insightful revelations into the impact of neuron type and the number of membership functions (MFs) on binary classification tasks. Table 1 encapsulates the accuracy, F-Score, recall, and precision metrics, affording us a comprehensive overview of the model's efficacy under different settings.

The results presented in Table 2 showcase the best performance metrics achieved in an experiment using a synthetic dataset designed to compare different levels of interpretability according to the increase in the number of membership functions. In this specific study, the primary variable was the type of logical neuron used, either andneuron or orneuron, along with a variation in the number of membership functions from 2 to 4. This configuration allowed for a detailed examination of how the varying complexity of the model's interpretative components influences its overall performance. The results summarized in the table represent the optimal outcomes obtained across all configurations, highlighting the effectiveness of the model under different interpretative settings. An initial analysis reveals significant performance disparities among neuron types, particularly in configurations with two (Fig. 9 and four Membership Functions (MFs). The configurations of orneuron and unineuron with four MFs achieved perfect scores across all metrics, underscoring the crucial role of an optimal number of MFs in boosting model precision and recall. This perfection in metrics suggests a sophisticated demarcation of decision boundaries, likely due to a nuanced partitioning of the feature space.

Table 2
Summary of Performance Metrics across Different Model Configurations-Synthetic dataset.

Neuron Type	MFs	Accuracy	F-Score	Recall	Precision
andneuron	2	0.433	0.605	1.000	0.433
andneuron	3	0.667	0.722	1.000	0.565
andneuron	4	0.667	0.722	1.000	0.565
orneuron	2	**1.000**	**1.000**	**1.000**	**1.000**
orneuron	3	**1.000**	**1.000**	**1.000**	**1.000**
orneuron	4	**1.000**	**1.000**	**1.000**	**1.000**
unineuron	2	0.867	0.867	1.000	0.765
unineuron	3	0.667	0.615	0.615	0.615
unineuron	4	0.867	0.867	1.000	0.765

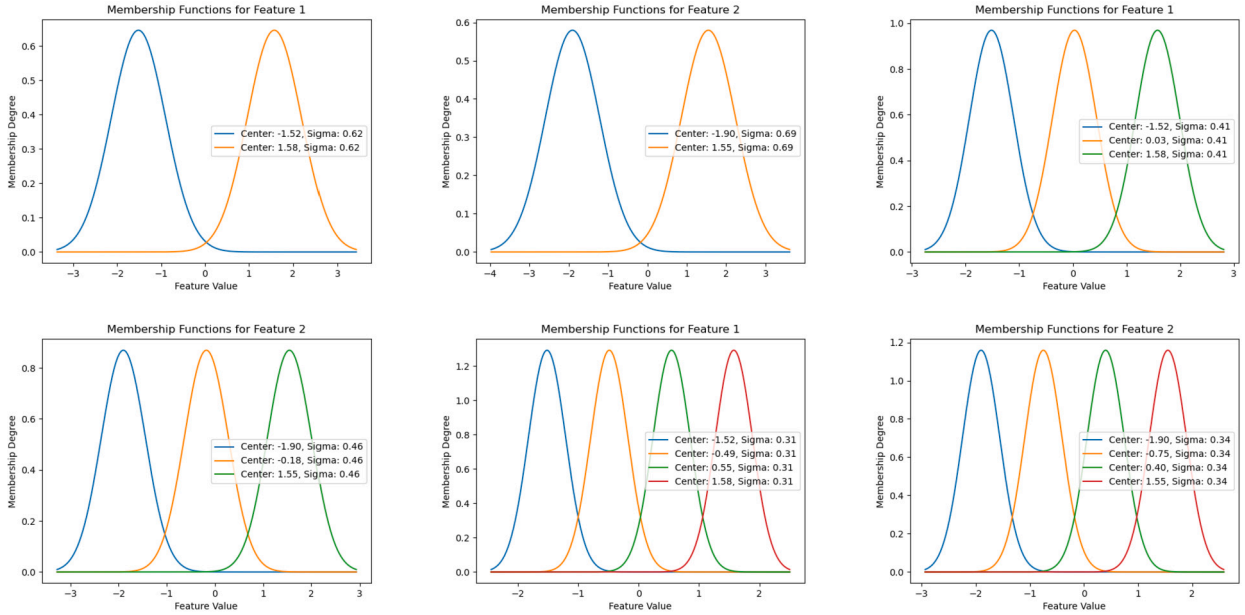


Fig. 9. Fuzzification process generated by the model in 2 dimensions.

Conversely, the andneuron configuration with three MFs showed a decline in performance, recording the lowest accuracy and precision. This could indicate a potential overcomplication within the andneuron model, where additional MFs do not necessarily correlate with improved performance. This might be attributed to an excessive saturation of the feature space, which could obstruct effective learning.

Orneuron models consistently outperformed their counterparts, showcasing resilience to the number of MFs, a characteristic less observed in other neuron types. This resilience may stem from orneuron's capability to identify crucial features and generalize across binary classification tasks, regardless of the MFs' granularity.

Unineuron models displayed a progressive enhancement in performance metrics as the number of MFs increased, suggesting a positive relationship between the number of MFs and the model's classification and recall accuracy. This indicates unineuron's adaptability and potential for fine-tuning towards peak performance.

These observations pave the way for an in-depth analysis of the models' interpretability, including similarities indicated in Fig. 12 (The redder, the greater the similarity. The less similar, the bluer the result), consistency (Fig. 13), where red indicates higher consistency values, this suggests that the rules or MFs compared are highly consistent with each other. Fig. 14 presents the same context with distinguishability. This means they tend to produce similar results or predictions. If the values are close to blue, it indicates problems in the consistency of those rules).

Analyzing Figs. 13 and 14, we can observe specific trends in the consistency and distinguishability of fuzzy rules across various configurations. In Fig. 14, the consistency matrix highlights the internal agreement between fuzzy rules under different logical neuron types (AndNeuron, OrNeuron, UniNeuron) and membership function setups (2MF, 3MF, 4MF). For instance, in the setup with 2MF and AndNeuron, rules show high consistency, particularly between rules 1 and 2, suggesting a strong alignment in their logical outcomes across tested scenarios.

In Fig. 14, the distinguishability matrix reveals differences in how distinctly each rule processes information. The 3MF setup with OrNeuron configuration demonstrates varying levels of distinguishability; for example, rule 3 is highly distinguishable from rules 0,

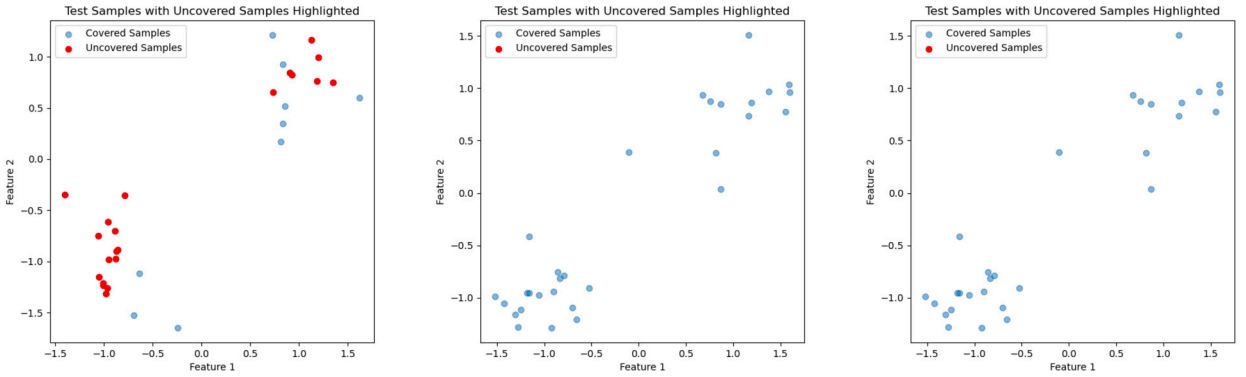


Fig. 10. ϵ-Completeness Matrix in the experiments (Each column represents AndNeuron, OrNeuron e Unineuron for 4MF.)

1, and 2, indicating it uniquely interprets input data compared to the others. This is crucial for ensuring that each rule contributes independently to the overall decision-making process, enhancing the interpretability and robustness of the model.

Finally, ϵ-completeness criteria (Fig. 10) indicates whether the membership functions adequately covered the samples during the training phase. It was observed that the andneuron model encountered significant issues with coverage throughout the experiment, elucidating its diminished performance. This aspect presents a compelling avenue for exploration in subsequent research endeavors. Understanding the interplay between model complexity, interpretability, and performance is essential, guiding toward configurations that not only achieve high accuracy but also maintain clarity and interpretability in decision-making processes.

6.2. State-of-the-art evaluation

This subsection articulates the outcomes of experiments that have been previously delineated. The findings from the thirty iterations conducted within the experiment are cataloged in Table 3, wherein the mean of the evaluated scores, along with the standard deviation encapsulated in parentheses, are presented.

Analyzing the IFNN model across various datasets reveals its performance dependency on the specific characteristics of the data, as detailed in the updated table (Table 3). On straightforward datasets like Iris, the model demonstrates exceptional performance, particularly the OrNeuron configuration, achieving perfect metrics. This highlights the model's compatibility with clear, distinct data patterns. However, challenges emerge with more complex datasets such as Haberman, underscoring the critical role of model configuration. The IFNN OrNeuron configuration outperforms others, emphasizing that selecting and fine-tuning the right model configuration is crucial for success.

The adaptability of the IFNN to diverse datasets such as Transfusion and Liver Data further illustrates the importance of matching model settings to dataset specifics to fully leverage FNN capabilities. The statistical analysis confirms the model's competitive edge in binary pattern classification (Table 4). For example, in "Liver Data," the variance in accuracy, precision, recall, and F1 Score across different model configurations reiterates the necessity of careful model configuration selection to maximize performance.

The statistical tests were applied using a blocked ANOVA approach to account for variations across different datasets, ensuring that the results reflected the performance of models independent of the dataset-specific characteristics. This method treats each dataset as a block, allowing us to isolate the effect of the models across all datasets and provide a robust assessment of their performance. The significance of the models' effects was evaluated, with post-hoc Tukey tests conducted for detailed pairwise comparisons when significant interactions were detected.

This comprehensive analysis, summarized in Tables 4, 5, and 6, highlights the versatility of the IFNN model in handling various data types and underscores the critical importance of model optimization to achieve optimal results in complex classification tasks.

The ANOVA results presented in Table 4 compare different models and datasets, showing significant interactions between models and datasets, indicating that the performance of each model varies depending on the dataset used. The P-values listed in the table show the probability by which the observed differences in model performance across datasets could occur by chance. Values such as 3.529262e-24 for the model effect and 1.132515e-53 for the dataset effect indicate that these factors significantly influence the outcomes at a confidence level usually above 95% or even 99%, underscoring the robustness of the findings.

Furthermore, Table 6 provides a detailed breakdown of pairwise comparisons between models using Tukey's HSD test, which is specifically designed to determine if the means of any pair of groups are statistically different while controlling for the type I error rate across multiple comparisons. This test is applied to all models over all datasets collectively to assess overall trends rather than individual dataset performance.

Shapiro-Wilk and Levene's tests, presented in Table 5, check for normal distribution of residuals and homogeneity of variances, respectively, which are prerequisites for the validity of ANOVA. The Shapiro-Wilk test result ($p < 0.05$) indicates non-normality in data, which may impact the ANOVA results, while the Levene's test ($p > 0.05$) suggests that variances are homogeneous across groups.

In Tukey's HSD test (Table 6), significant p-values (e.g., 0.027 between FNN AndNeuron and FNN OrNeuron) indicate that the performance differences are statistically significant, whereas non-significant results (e.g., 0.693 between FNN AndNeuron and FNN

Table 3
Results of the experiments.

Model	Accuracy	Precision	Recall	F1 Score
Dataset: Iris				
IFNN AndNeuron	0.79 (0.21)	0.78 (0.31)	0.91 (0.12)	0.78 (0.12)
IFNN OrNeuron	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)
IFNN UniNeuron	0.99 (0.01)	1.00 (0.00)	0.98 (0.02)	0.99 (0.01)
Random Forest	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)
MLP	0.88 (0.11)	0.63 (0.45)	0.65 (0.45)	0.64 (0.45)
Naive Bayes	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)
Dataset: Mammographic				
IFNN AndNeuron	0.67 (0.04)	0.64 (0.11)	0.82 (0.18)	0.69 (0.02)
IFNN OrNeuron	0.82 (0.05)	0.81 (0.05)	0.81 (0.04)	0.81 (0.04)
IFNN UniNeuron	0.68 (0.01)	0.64 (0.02)	0.71 (0.02)	0.67 (0.00)
Random Forest	0.79 (0.03)	0.77 (0.04)	0.78 (0.02)	0.78 (0.03)
MLP	0.81 (0.03)	0.81 (0.05)	0.77 (0.10)	0.78 (0.04)
Naive Bayes	0.81 (0.04)	0.77 (0.05)	0.83 (0.02)	0.80 (0.04)
Dataset: Haberman				
IFNN AndNeuron	0.50 (0.10)	0.83 (0.02)	0.42 (0.15)	0.55 (0.14)
IFNN OrNeuron	0.70 (0.01)	0.86 (0.01)	0.73 (0.01)	0.79 (0.01)
IFNN UniNeuron	0.56 (0.05)	0.79 (0.04)	0.57 (0.03)	0.66 (0.04)
Random Forest	0.70 (0.02)	0.79 (0.00)	0.82 (0.02)	0.81 (0.01)
MLP	0.76 (0.04)	0.78 (0.01)	0.96 (0.07)	0.86 (0.03)
Naive Bayes	0.78 (0.00)	0.79 (0.00)	0.97 (0.00)	0.87 (0.00)
Dataset: Transfusion				
IFNN AndNeuron	0.78 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
IFNN OrNeuron	0.77 (0.00)	0.47 (0.01)	0.25 (0.03)	0.32 (0.02)
IFNN UniNeuron	0.76 (0.01)	0.33 (0.01)	0.08 (0.04)	0.12 (0.05)
Random Forest	0.68 (0.00)	0.29 (0.01)	0.28 (0.00)	0.28 (0.00)
MLP	0.78 (0.00)	0.51 (0.01)	0.31 (0.01)	0.39 (0.01)
Naive Bayes	0.78 (0.00)	0.50 (0.00)	0.16 (0.00)	0.24 (0.00)
Dataset: Liver Data				
IFNN AndNeuron	0.54 (0.03)	0.49 (0.03)	0.56 (0.20)	0.50 (0.12)
IFNN UniNeuron	0.71 (0.01)	0.66 (0.02)	0.72 (0.02)	0.69 (0.01)
FNN UniNeuron	0.64 (0.02)	0.60 (0.03)	0.53 (0.05)	0.56 (0.04)
Random Forest	0.70 (0.02)	0.71 (0.03)	0.54 (0.02)	0.61 (0.02)
MLP	0.60 (0.05)	0.66 (0.18)	0.33 (0.18)	0.39 (0.14)
Naive Bayes	0.60 (0.00)	0.53 (0.00)	0.70 (0.00)	0.60 (0.00)

Table 4
ANOVA Results.

Source	sum_sq	df	F	PR(>F)
C(Model)	0.494669	5	40.378497	3.529262e-24
C(Dataset)	2.102787	4	214.556173	1.132515e-53
C(Model):C(Dataset)	0.379908	20	7.752725	8.922679e-14
Residual	0.294019	120	NaN	NaN

Table 5
Shapiro-Wilk and Levene's Test Results.

Test	Statistics	p-value
Shapiro-Wilk	0.755	1.49e-14
Levene's	1.381	0.235

UniNeuron) suggest no significant difference in performance between those models. These outcomes guide the selection of models for specific scenarios based on empirical evidence and performance metrics, ensuring a well-informed choice in deployment contexts.

Tukey's HSD test, in particular, helps identify models that perform statistically similarly under certain conditions, as indicated by the high p-Tukey values in Table 6, where models with non-significant differences in mean performance are highlighted in gray. This analysis allows for an easier comparison between models where significant performance disparities do not exist, thereby facilitating the identification of models with equivalent efficacy. Such insights underscore the importance of a judicious selection of models based on both statistical evidence and practical considerations to ensure the best possible outcomes in classification tasks. The validation assumptions highlighted in Table 5 through Shapiro-Wilk and Levene's tests further corroborate the findings, suggesting that while some IFNN models may offer similar performance in certain scenarios, the significant performance differences observed in others emphasize the role of aligning model choices with the specific characteristics of datasets.

Table 6
Tukey's HSD Test Results.

A	B	mean(A)	mean(B)	diff	se	T	p-tukey	hedges
FNN AndNeuron	FNN OrNeuron	0.593	0.772	-0.179	0.056	-3.148	0.027	-1.046
FNN AndNeuron	FNN UniNeuron	0.593	0.675	-0.082	0.056	-1.456	0.693	-0.443
FNN AndNeuron	MLP	0.593	0.742	-0.149	0.056	-2.626	0.102	-0.863
FNN AndNeuron	Naive Bayes	0.593	0.767	-0.174	0.056	-3.065	0.034	-0.958
FNN AndNeuron	Random Forest	0.593	0.772	-0.179	0.056	-3.573	0.008	-1.263
FNN OrNeuron	FNN UniNeuron	0.772	0.675	0.097	0.056	1.692	0.541	0.597
FNN OrNeuron	MLP	0.772	0.742	0.030	0.056	0.522	0.995	0.204
FNN OrNeuron	Naive Bayes	0.772	0.767	0.005	0.056	0.083	1.000	0.030
FNN OrNeuron	Random Forest	0.772	0.772	0.000	0.056	-0.424	0.998	-0.185
FNN UniNeuron	MLP	0.675	0.742	-0.067	0.056	-1.170	0.850	-0.407
FNN UniNeuron	Naive Bayes	0.675	0.767	-0.092	0.056	-1.609	0.595	-0.530
FNN UniNeuron	Random Forest	0.675	0.772	-0.097	0.056	-2.117	0.289	-0.801

Table 7
Performance metrics of models on the sepsis dataset.

Model	Accuracy	Precision	Recall	F1 Score
FNN AndNeuron	0.927	0.962	1.000	0.927
FNN OrNeuron	0.927	0.927	1.000	0.962
FNN UniNeuron	0.927	0.926	1.000	0.962
Random Forest	0.926	0.926	1.000	0.962
MLP	0.926	0.926	1.000	0.962
Naive Bayes	0.926	0.926	1.000	0.962
SVM	0.926	0.926	1.000	0.962
Logistic Regression	0.926	0.926	1.000	0.962

Statistical evaluations using ANOVA and Tukey's HSD tests reveal that the performance of FNN models varies significantly with the dataset applied, illustrating the context-dependent effectiveness of these models. The interaction effect noted in the ANOVA analysis underscores that no single model consistently outperforms others across different datasets. Further analyses validate these findings, with Tukey's HSD highlighting that while some IFNN models perform similarly in certain contexts, suggesting flexibility in model selection. Significant performance differences in others stress the importance of choosing models aligned with specific dataset characteristics. This emphasizes the necessity of careful model selection to achieve optimal classification results, guided by both statistical evidence and practical considerations.

6.3. Analysis of the sepsis identification results

In this section, we present the comprehensive evaluation results of our experiments conducted on the sepsis dataset. For clarity and uniformity in the model's interpretive framework, each feature was fuzzified using two fuzzy sets. For example, the "Age" feature might use linguistic terms such as 'Young' and 'Old', while "Sex" could be represented as 'Male' and 'Female'. This approach not only aligns with common interpretative practices but also facilitates easier understanding and communication of the model's reasoning processes. This decision was based on preliminary tests that evaluated the model's performance with two, three, and four membership functions (MF), where two MFs provided the best balance between interpretability and accuracy.[2]

The performance of three variants of the Adam-optimized Fuzzy Neural Network (FNN) models—AndNeuron, OrNeuron, and UniNeuron—is compared with state-of-the-art machine learning models including Random Forest, Multi-Layer Perceptron (MLP), Naive Bayes, Support Vector Machine (SVM), and Logistic Regression. The accuracy, precision, recall, and F1 score achieved by each model are displayed in Table 7.

Furthermore, the fuzzy sets' semantics help in contextualizing the model's classifications. For instance, 'MF1' and 'MF2' in the context of 'Age' could be interpreted as 'Young' (low range) and 'Old' (high range), respectively. Similarly, for 'Sex', 'MF1' might correspond to 'Male' and 'MF2' to 'Female'. This semantic labeling aids in understanding the rationale behind each rule generated by the model, as shown in Tables 7 and 8. The use of different shades of blue and red in the tables emphasizes the certainty of the consequents, with deeper colors indicating higher confidence in the survival or non-survival prediction, respectively.

The analysis of Adam-optimized Fuzzy Neural Network (FNN) models on a sepsis dataset demonstrates their competitive performance with perfect recall and balanced accuracy, precision, and F1 scores, highlighting their applicability in clinical settings by effectively identifying all sepsis cases. The fuzzification process enhances interpretability, illustrated in Fig. 11 and elaborated through fuzzy rules in Tables 8, 9, and 10, bridging model outputs with human understanding. Notably, the model's streamlined architecture, consisting of only 9 fuzzy rules, indicates an efficient model design without sacrificing performance. This positioning is alongside state-of-the-art models in medical applications.

[2] After previous test using MF = 2,3,4.

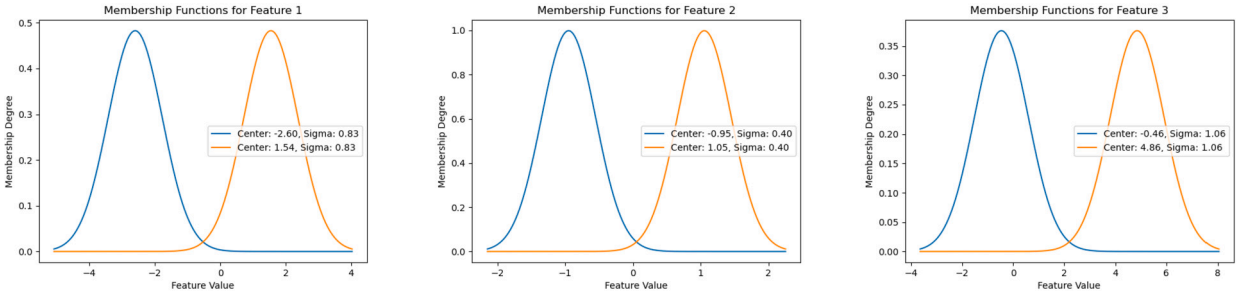


Fig. 11. Fuzzification process generated by the model in 2 dimensions in Sepsis dataset.

Table 8
Fuzzy Rules Generated by AndNeuron. Note: Parentheses indicate the impact on the Membership Function (MF).

Rule	Conseq.	Situation
IF Age is MF1 (0.67) AND Sex is MF1 (0.58) AND Ep. number is MF1 (0.25)	13.30	Alive
IF Age is MF1 (0.54) AND Sex is MF1 (0.33) AND Ep. number is MF2 (0.12)	2.58	Alive
IF Age is MF1 (0.63) AND Sex is MF2 (0.80) AND Ep. number is MF1 (0.58)	16.79	Alive
IF Age is MF1 (0.44) AND Sex is MF2 (0.74) AND Ep. number is MF2 (0.45)	2.59	Alive
IF Age is MF2 (0.14) AND Sex is MF1 (0.51) AND Ep. number is MF1 (0.70)	-3.95	Deceased
IF Age is MF2 (0.53) AND Sex is MF1 (0.48) AND Ep. number is MF2 (0.98)	0.53	Alive
IF Age is MF2 (0.07) AND Sex is MF2 (0.88) AND Ep. number is MF1 (1.00)	-2.86	Deceased
IF Age is MF2 (0.27) AND Sex is MF2 (0.72) AND Ep. number is MF2 (0.59)	1.12	Alive

Table 9
Fuzzy Rules Generated by OrNeuron. Note: Parentheses indicate the impact on the Membership Function (MF).

Rule	Conseq.	Situation
IF Age is MF1 (0.49) OR Sex is MF1 (0.14) OR Ep. number is MF1 (0.12)	0.89	Alive
IF Age is MF1 (0.07) OR Sex is MF1 (0.73) OR Ep. number is MF2 (0.93)	1.36	Alive
IF Age is MF1 (0.15) OR Sex is MF2 (0.57) OR Ep. number is MF1 (0.82)	1.30	Alive
IF Age is MF1 (0.13) OR Sex is MF2 (0.85) OR Ep. number is MF2 (0.57)	5.48	Alive
IF Age is MF2 (0.40) OR Sex is MF1 (0.76) OR Ep. number is MF1 (0.66)	0.11	Alive
IF Age is MF2 (0.33) OR Sex is MF1 (0.47) OR Ep. number is MF2 (0.41)	-0.52	Deceased
IF Age is MF2 (0.89) OR Sex is MF2 (0.40) OR Ep. number is MF1 (0.59)	-0.35	Deceased
IF Age is MF2 (0.35) OR Sex is MF2 (0.44) OR Ep. number is MF2 (0.09)	-1.49	Deceased

Table 10
Fuzzy Rules Generated by UniNeuron. Note: Parentheses indicate the impact on the Membership Function (MF).

Rule	Conseq.	Situation
IF Age is MF1 (0.61) OR Sex is MF1 (0.33) OR Ep. number is MF1 (0.14)	0.27	Alive
IF Age is MF1 (0.31) AND Sex is MF1 (0.83) AND Ep. number is MF2 (0.90)	-0.06	Deceased
IF Age is MF1 (0.20) AND Sex is MF2 (0.55) AND Ep. number is MF1 (0.91)	0.25	Alive
IF Age is MF1 (0.72) AND Sex is MF2 (0.73) AND Ep. number is MF2 (0.44)	0.52	Alive
IF Age is MF2 (0.57) OR Sex is MF1 (0.33) OR Ep. number is MF1 (0.40)	-0.02	Imprecise
IF Age is MF2 (0.70) AND Sex is MF1 (0.32) AND Ep. number is MF2 (0.41)	-0.02	Imprecise
IF Age is MF2 (0.41) OR Sex is MF2 (0.71) OR Ep. number is MF1 (0.94)	-0.33	Deceased
IF Age is MF2 (0.03) AND Sex is MF2 (0.55) AND Ep. number is MF2 (0.67)	0.01	Imprecise

Note: In the rules below (Table 8), the numbers in parentheses next to each membership function (MF) represent the impact of that MF. In the last column the blue colors indicate that the patient survived and the red colors indicate that the patient didn't survive.

In Table 10, the class "Imprecise" emerges as a significant category within the outcomes of the IFNN with UniNeuron-generated rules. This classification is utilized when the algorithm determines that the available evidence is insufficient to make a reliable prediction towards either surviving ('Alive') or not surviving ('Deceased'). Specifically, the "Imprecise" label is applied to outcomes where the consequents of fuzzy rules are near zero, indicating a balanced or neutral impact from the rules' conditions, leading to ambiguous decision outputs. The introduction of an "Imprecise" class is a key feature of our model, aligning with the demand for explanatory algorithms that inform the user when the decision-making process does not yield a clear result. This ability to flag

uncertainty enhances the transparency of the model, providing a useful alert mechanism that highlights the need for further data or review before making a critical decision.

Analyzing “Age,” “Sex,” and “Number of Episodes” through IFNN models reveals their critical impact on sepsis prognosis, improving interpretability with consistent risk representation and highlighting prediction complexities. Despite outcome variations indicating the nuanced application of models in clinical settings, these rules provide a solid base for sepsis management with accuracy over 92%. Visual assessments of rule similarity, consistency, and ϵ-completeness in Figs. 15, 16, and 17 further refine this analysis. Specifically, inconsistencies noted in Rule 4 for the OrNeuron suggest avenues for analytical improvements, emphasizing the careful integration of model insights for enhanced patient outcomes.

7. Conclusion

The IFNN model introduces a novel blend of fuzzy logic and neural networks, aiming for a sweet spot between interpretability and performance. With the Adam optimization algorithm at its core, the model showcases enhanced accuracy across diverse datasets, setting a benchmark for AI systems where clarity does not come at the expense of precision.

7.1. Achievements

This study highlights the model’s effectiveness in deriving data-driven insights through statistical analysis, interpretability validations, and practical relevance, particularly in transforming logical neurons into explicit fuzzy rules for transparent AI decision-making. The model’s application to sepsis case studies showcases its compatibility with leading performances and ability to distill complex data into a simplified rule set, which is crucial for healthcare and enhances the general accessibility of AI-supported decisions. This approach fosters rich, interpretable narratives, emphasizing the model’s contribution to making AI reasoning processes more comprehensible.

Compared to state-of-the-art models like MLP, SVM, and Random Forests, IFNN has demonstrated superior interpretability without compromising on performance metrics such as accuracy, precision, and recall. These achievements justify the proposed method as a viable solution to the problem of opaque decision-making in AI, particularly in critical fields like healthcare.

7.2. Challenges and future directions

The IFNN model, while showcasing strengths in accuracy and interpretability, encounters difficulties with large sample sizes and high-dimensional datasets due to its equally spaced membership functions, impacting scalability and efficiency. Future directions will focus on enhancing the model’s capability to manage extensive datasets and complex multidimensional challenges, likely employing pruning techniques to simplify the model’s structure and improved scalability. Furthermore, advancements in interpretability methods will be prioritized to maintain the model’s transparency and usability. As it stands, IFNN represents a significant step towards integrating precision with clarity in AI solutions, yet its progression hinges on overcoming scalability and high-dimensional data challenges to fully realize its potential and inspire further developments in the field of interpretable AI.

Further research will explore the integration of more dynamic fuzzy set generation techniques to handle varied data distributions and improve the model’s adaptability. Additionally, expanding the repertoire of fuzzy logical operators and integrating machine learning advancements could enhance both the depth and breadth of interpretability in future versions of IFNN. This future work will strive to solidify the model’s position as a pioneering solution in the interpretability-focused landscape of artificial intelligence.

Moreover, plans include adapting the model for evolving environments, allowing for the dynamic creation of fuzzy neurons in response to changing data streams. Improvements in optimization algorithms like Particle Swarm Optimization (PSO) are also future options to refine learning processes further. Finally, studies will be conducted to extend the model’s capabilities to multi-class classification scenarios, broadening its applicability across different fields.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

CRediT authorship contribution statement

Paulo Vitor de Campos Souza: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **Mauro Dragoni:** Writing – review & editing, Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Paulo Vitor de Campos Souza reports financial support was provided by Bruno Kessler Foundation. Paulo Vitor de Campos Souza reports a relationship with Bruno Kessler Foundation that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT 3.5 to improve reading and text fluency. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Appendix A

A.1. Interpretability matrices from synthetic dataset

This appendix presents the graphical results obtained by Adapt-IFNN in the evaluation of the synthetic data listed in Section 6.1.

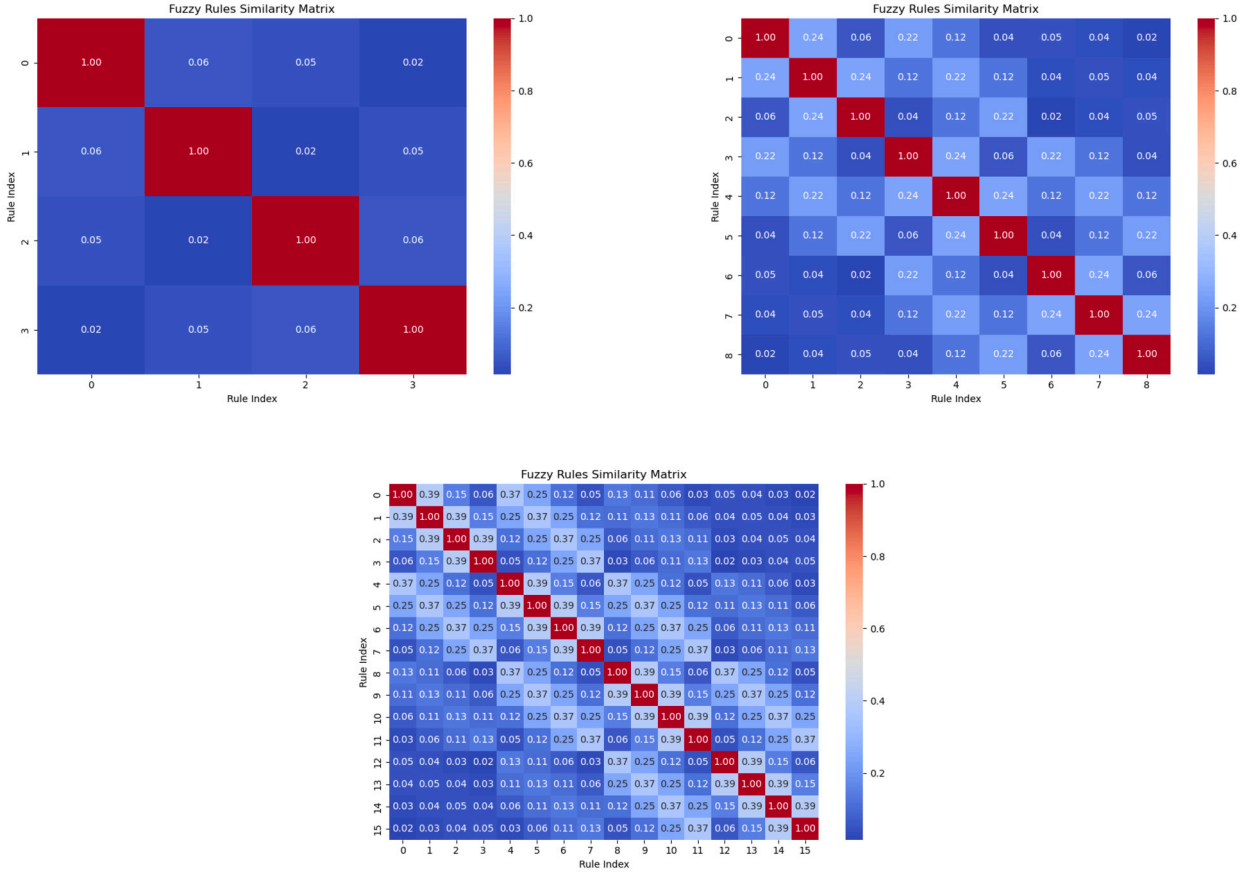


Fig. 12. Similarity Matrix in the experiments (Each column represents AndNeuron, OrNeuron e Unineuron for 2MF, 3MF and 4MF, respectively.)

A.2. Sepsis dataset

This appendix contains the similarity, distinguishability matrices and e-completeness criteria of the tests carried out with the sepsis dataset.

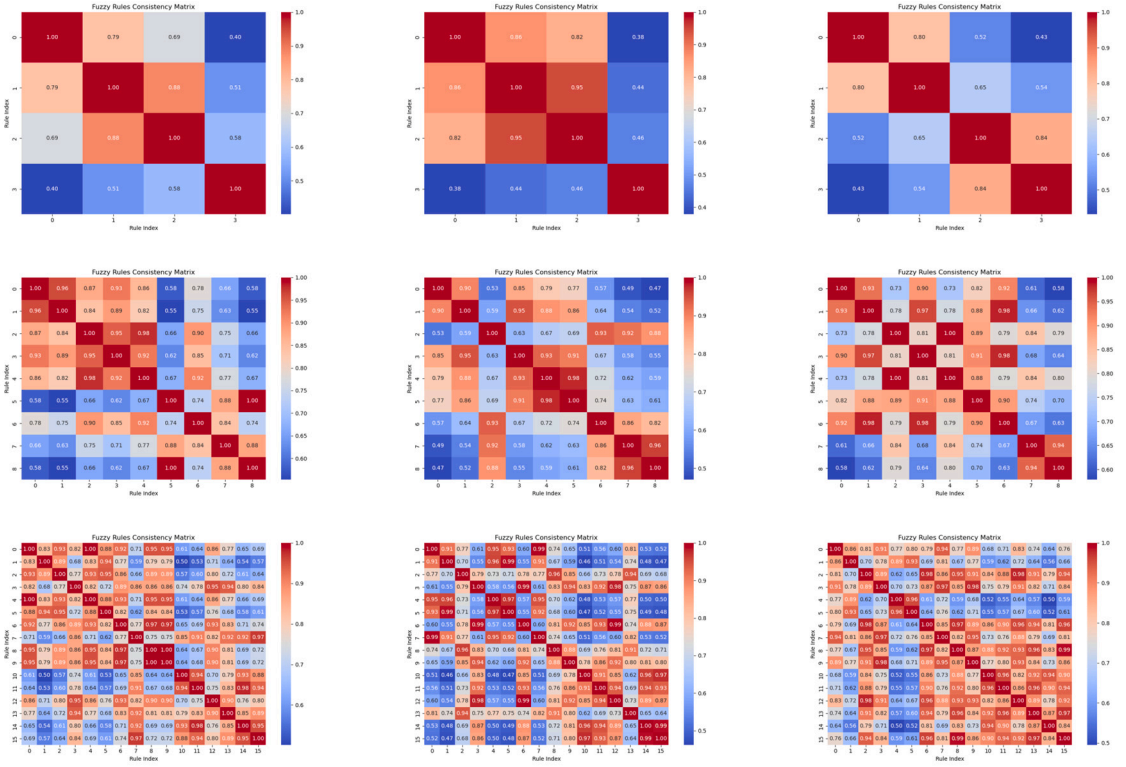


Fig. 13. Consistency Matrix in the experiments (Each column represents AndNeuron, OrNeuron e Unineuron for 2MF, 3MF and 4MF, respectively.)

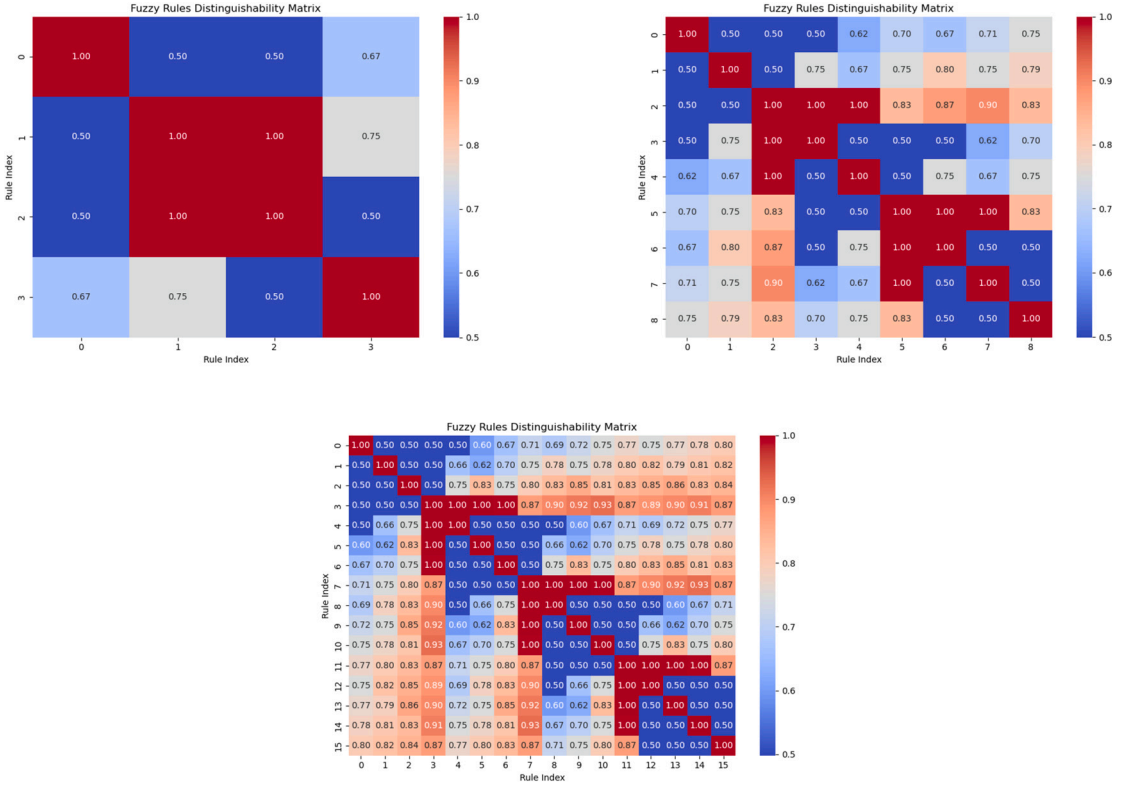


Fig. 14. Distinguishability Matrix in the experiments; AndNeuron, OrNeuron e Unineuron for 2MF, 3MF and 4MF, respectively).

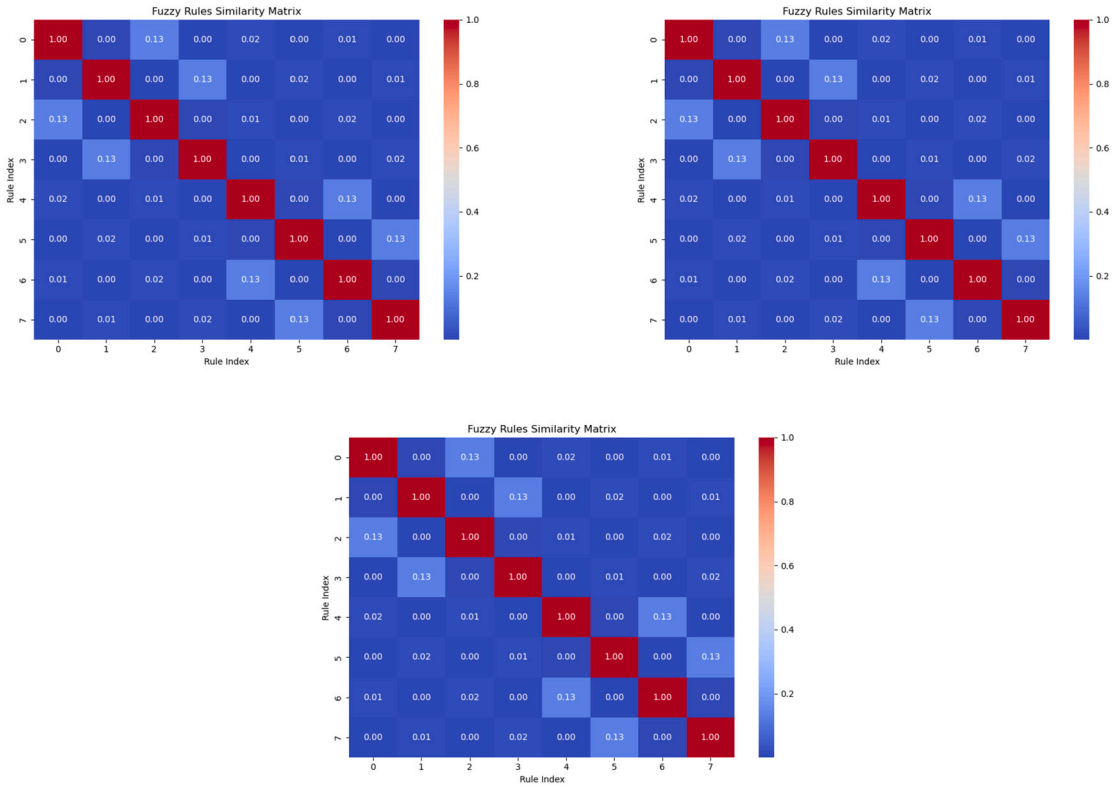


Fig. 15. Similarities of fuzzy rules- Sepsis dataset.

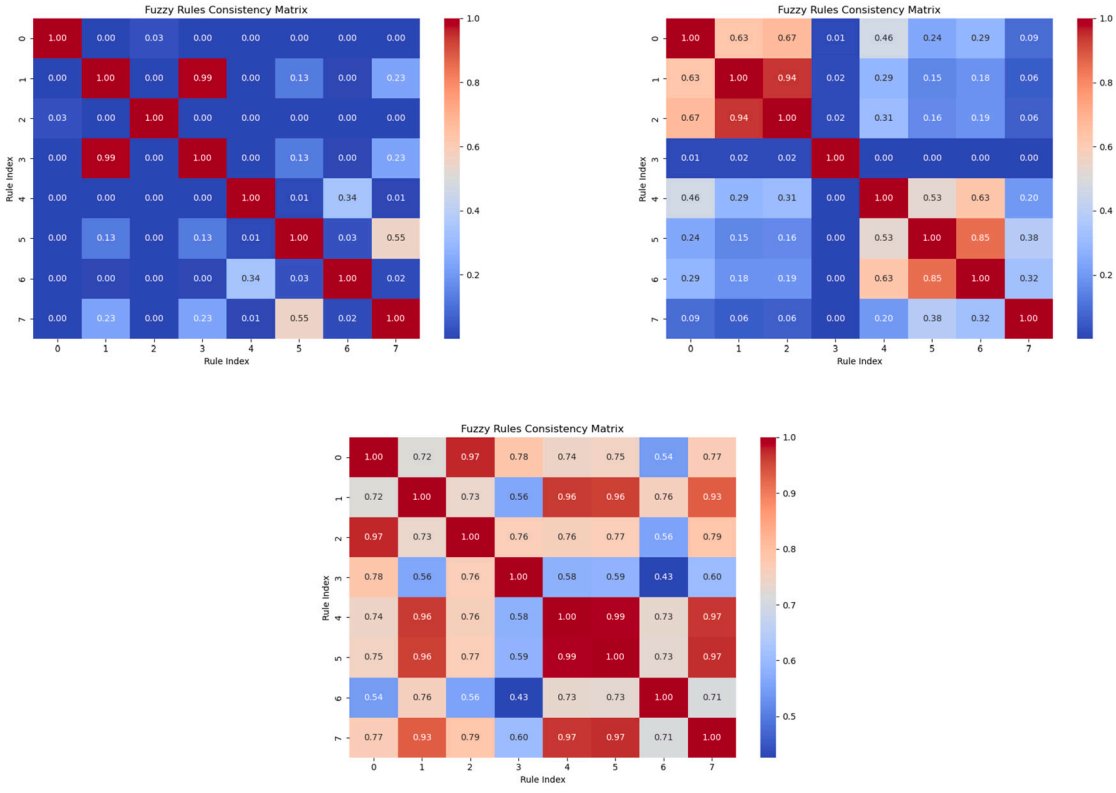


Fig. 16. Consistencies of fuzzy rules- Sepsis dataset.

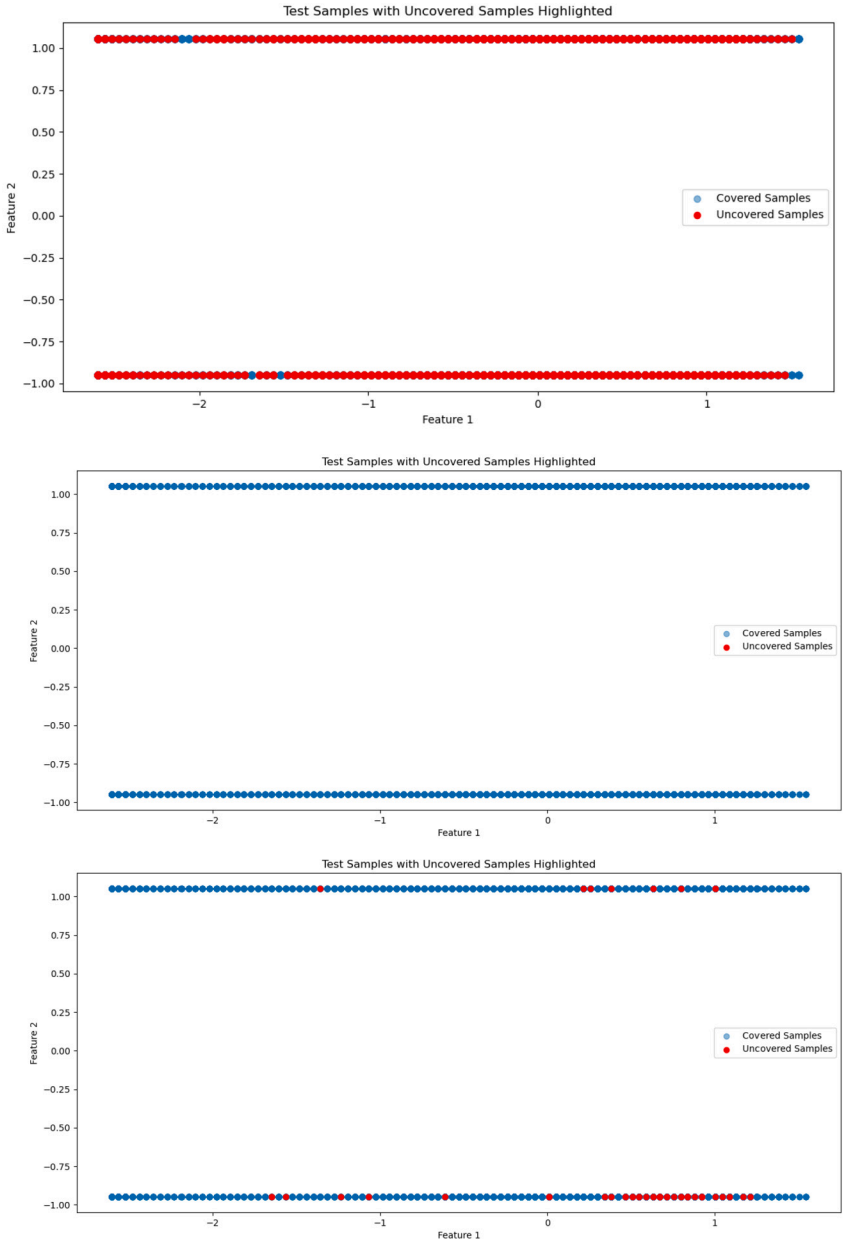


Fig. 17. Similarities of fuzzy rules- Sepsis dataset (andneuron, Orneuron and Unineuron respectively.

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